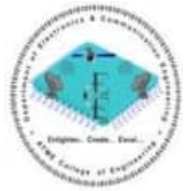




A T M E
College of Engineering



Subject Name: Microwave Engineering & Antenna Theory

Subject Code: BEC701

Course Coordinator:

Guruprasad K N

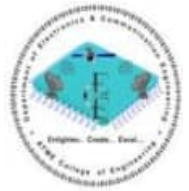
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Module 2

Microwave Network Theory & Microwave Passive Devices

Module 2: Chapter I

Microwave Network Theory

- Introduction
- Symmetrical Z and Y-Parameters for Reciprocal Networks
- S matrix representation of Multi-Port Networks

Suggested Reading

Text I: “Microwave Engineering” -
Annapurna Das, Sisir K Das, TMH Publication, 2nd Edition,
2010,

Text I:

Sections: 6.1, 6.2, 6.3

□ Introduction :

A microwave network consists of coupling of various microwave components and devices such as attenuators, phase shifters, amplifiers, resonators etc., to sources through transmission lines or waveguides. Connection of two or more microwave devices and components to a single point results in a *microwave junction*.

In a low frequency network, the input and output variables are voltage and current which can be related in terms of impedance Z-parameters, or admittance Y-parameters or hybrid h-parameters or ABCD parameters. These relationships for a two-port network of figure 4.1 can be represented by

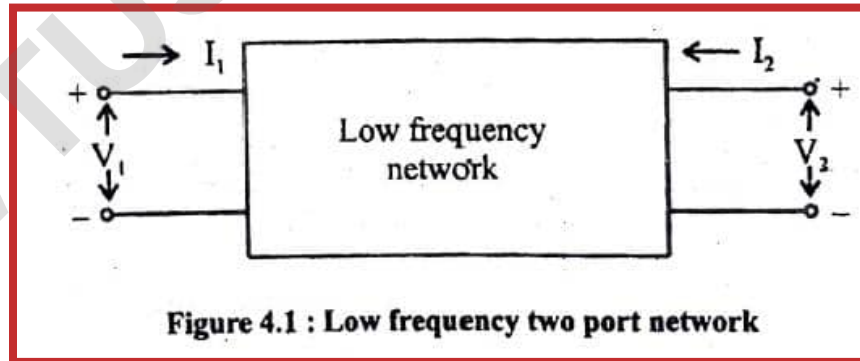


Figure 4.1 : Low frequency two port network

Impedance Z-parameter :

$$\left. \begin{aligned} V_1 &= Z_{11} I_1 + Z_{12} I_2 \\ V_2 &= Z_{21} I_1 + Z_{22} I_2 \end{aligned} \right\} \quad \dots (4.1)$$

or

$$\begin{bmatrix} V_1 \\ V_2 \end{bmatrix} = \begin{bmatrix} Z_{11} & Z_{12} \\ Z_{21} & Z_{22} \end{bmatrix} \begin{bmatrix} I_1 \\ I_2 \end{bmatrix} \quad \dots (4.2)$$

Admittance Y-parameter :

$$\left. \begin{aligned} I_1 &= Y_{11} V_1 + Y_{12} V_2 \\ I_2 &= Y_{21} V_1 + Y_{22} V_2 \end{aligned} \right\} \quad \dots (4.3)$$

or

$$\begin{bmatrix} I_1 \\ I_2 \end{bmatrix} = \begin{bmatrix} Y_{11} & Y_{12} \\ Y_{21} & Y_{22} \end{bmatrix} \begin{bmatrix} V_1 \\ V_2 \end{bmatrix} \quad \dots (4.4)$$

Hybrid h-parameter :

$$\left. \begin{aligned} V_1 &= h_{11} I_1 + h_{12} V_2 \\ I_2 &= h_{21} I_1 + h_{22} V_2 \end{aligned} \right\} \quad \text{..... (4.5)}$$

or

$$\begin{bmatrix} V_1 \\ I_2 \end{bmatrix} = \begin{bmatrix} h_{11} & h_{12} \\ h_{21} & h_{22} \end{bmatrix} \begin{bmatrix} I_1 \\ V_2 \end{bmatrix} \quad \text{..... (4.6)}$$

ABCD parameter :

$$\left. \begin{aligned} V_1 &= AV_2 - BI_2 \\ I_1 &= CV_2 - DI_2 \end{aligned} \right\} \quad \text{..... (4.7)}$$

or

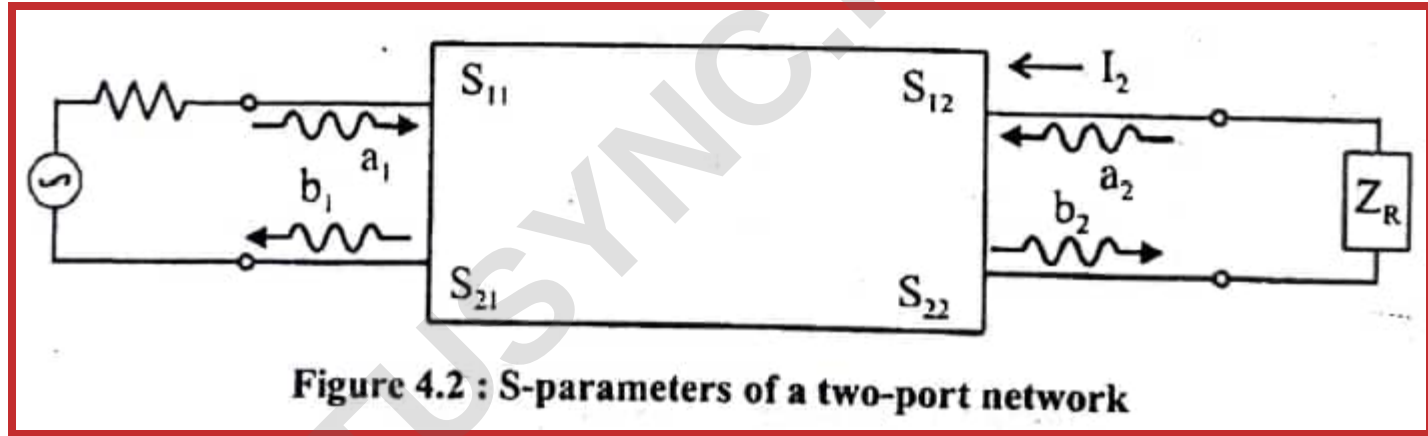
$$\begin{bmatrix} V_1 \\ I_1 \end{bmatrix} = \begin{bmatrix} A & -B \\ C & -D \end{bmatrix} \begin{bmatrix} V_2 \\ I_2 \end{bmatrix} \quad \text{..... (4.8)}$$

These parameters, Z , Y , h and ABCD parameters can be easily measured at low frequencies under short or open circuit conditions and can be used for analyzing the circuit.

The physical length of the device or the line at microwave frequencies, is comparable to or much larger than the wavelength. Due to this, the voltage and current are difficult to measure as also the above mentioned parameters. The reasons for this are listed as below.

- (a) Equipment is not available to measure the total voltage and total current at any point.
- (b) Over a wide range of frequencies, short and open circuits are difficult to realize.
- (c) Active devices such as power transistors, tunnel diodes etc, will become unstable under short or open circuit conditions.

Therefore, a new representation is needed to overcome these problems at microwave frequencies. The logical variables are travelling waves rather than voltages and currents and these variables are labelled as “**Scattering or S-parameters**”. These parameters for a two-port network are represented as shown in figure 4.2.



These S-parameters can be represented in an equation form related to the travelling waves a_1 , a_2 , b_1 and b_2 through

$$\begin{cases} b_1 = S_{11} a_1 + S_{12} a_2 \\ b_2 = S_{21} a_1 + S_{22} a_2 \end{cases} \quad \text{..... (4.9)}$$

Reciprocal Network:

A reciprocal network is one in which the transmission of a signal between any two ports does not depend on direction of propagation i.e. input and output ports are interchangeable.

(Eg: Scattering Parameter $S_{21} = S_{12}$, $S_{13} = S_{31}$)

Symmetrical Z and Y-Parameters for Reciprocal Networks:

In a reciprocal network, the junction media are characterized by scalar electrical parameters namely absolute permeability μ and absolute permittivity ϵ . In such a network, the impedance and the admittance matrices become symmetrical. This property can be proved by considering an N-port network.

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Seventh Semester B.E. Degree Examination, June/July 2019
Microwave and Antennas

Time: 3 hrs.

Max. Marks: 80

Note: Answer any FIVE full questions, choosing ONE full question from each module.

Module-1

1.
 - a. Discuss mechanism of oscillation in Reflex Klystron with schematic. (06 Marks)
 - b. A Reflex Klystron is to be operated at 10GHz with dc beam voltage 300V, repeller space 0.1cm for $1\frac{3}{4}$ mode. Calculate P_{RTOX} and corresponding repeller voltage for a beam current of 20mA. (05 Marks)
 - c. A transmission line has the following parameters: $R = 2\Omega/\text{m}$, $G = 0.5\text{mho}/\text{m}$, $f = 1\text{GHz}$, $L = 8\text{nH}/\text{m}$ and $C = 0.23\text{PF}$. Calculate its characteristic impedance and propagation constant. (05 Marks)

OR

2.
 - a. A line of 400Ω is connected to a load of $200 + j300\Omega$ which is excited by a matched generator at 800MHz. Find the location and length of a single stub nearest to the load to produce an impedance match. (08 Marks)
 - b. A certain transmission line has a characteristic impedance of $75 + j0.01\Omega$ and is terminated in a load impedance of $75 + j50\Omega$. Compute: i) Reflection coefficient ii) The transmission coefficient. (04 Marks)
 - c. What are the high frequency limitations of conventional vacuum tube / transistors? (04 Marks)

Module-2

3.
 - a. Show that impedance and admittance matrices are symmetrical for a reciprocal junction. (06 Marks)

8. compulsorily draw diagonal cross lines on the remaining blank pages, on, appeal to evaluator and on signatures written on pg. 42-8-50, will be treated as malpractice.

Let E_i and H_i be the respective electric and magnetic field intensities at the i^{th} port and let the total voltage $V_n = 0$ at all ports for $n = 0, 1, 2, \dots$ except at i^{th} port. Similarly if E_j and H_j are considered for the j^{th} port with $V_n = 0$ at other ports

Then from Lorentz reciprocity Theorem, We have

$$\oint_S (E_i \times H_j - E_j \times H_i) \cdot dS = 0 \quad \dots (4.10)$$

where S is the closed surface area including the junction and all the N number of ports. Since equation (4.10) holds good for all pairs of ports, we must have

$$\sum_{n=1}^N \oint_{t_n} [(E_i \times H_j) - (E_j \times H_i)] \cdot dS = 0 \quad \dots (4.11)$$

Since all V_n except V_i and V_j are zero, equation (4.11) reduces to

$$\oint_{t_i} (E_i \times H_j) \cdot dS - \oint_{t_j} (E_j \times H_i) \cdot dS = 0 \quad \dots (4.12)$$

Using Poynting theorem,

$$P_{ij} - P_{ji} = 0$$

$$\text{or } P_{ij} = P_{ji}$$

..... (4.13)

where P_{ij} = power at reference plane i due to input at plane j

and P_{ji} = power at reference plane j due to input at plane i

From equation (4.13)

$$P_{ij} = P_{ji}$$

$\therefore$

$$V_i I_{ij} = V_i I_{ji}$$

$$V_j [Y_{ij} V_i] = V_i [Y_{ji} V_j]$$

$\therefore$

$$Y_{ij} = Y_{ji}$$

..... (4.14)

It follows that

$$Z_{ij} = Z_{ji}$$

..... (4.15)

These two equations prove that the impedance and the admittance matrices are symmetrical for a reciprocal junction.

S matrix representation of Multi-Port Networks:

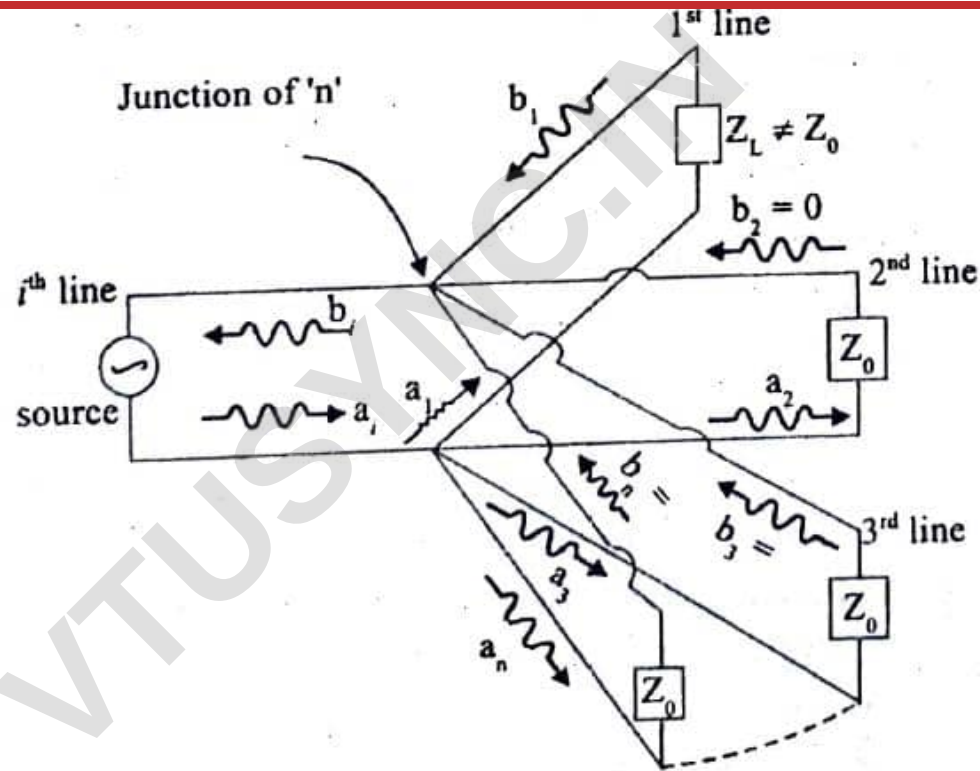
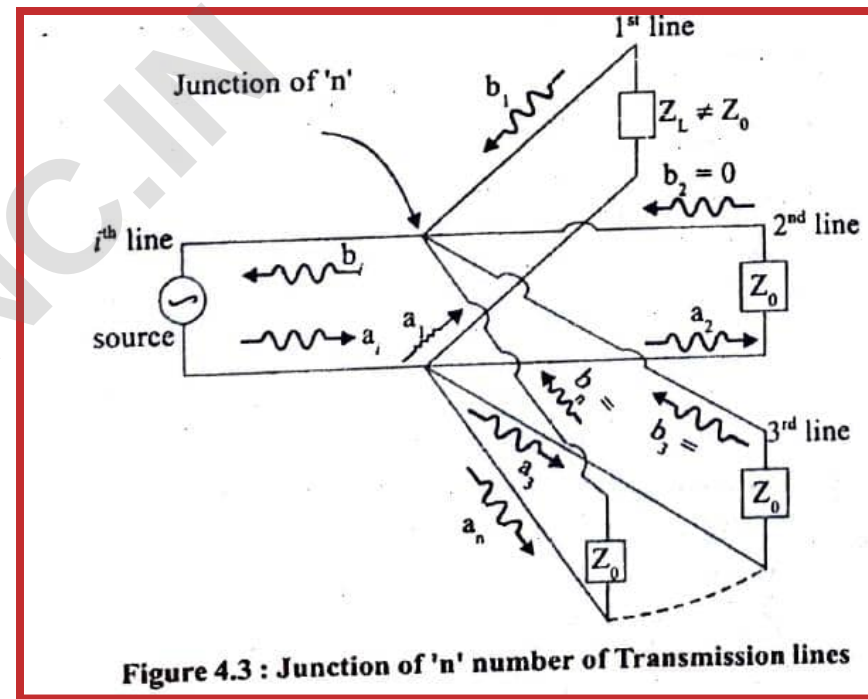


Figure 4.3 : Junction of ' n ' number of Transmission lines

In order to understand the significance of scattering matrix, let us consider a junction of "n" number of transmission lines as shown in figure 4.3. Let the i^{th} line (i can be any line from 1 to n) be terminated in a source, the first line in an impedance other than its characteristic impedance Z_0 and the remaining lines from 2nd to n^{th} lines in impedances equal to Z_0 . Let a_i be the incident wave (either voltage wave or current wave) at the junction due to the source connected at the i^{th} line. This divides itself among the $(n-1)$ number of transmission lines with a_1 going into 1st line, a_2 into 2nd line, a_3 into 3rd line and a_n into n^{th} line.



Since the transmission lines from 2 to n are terminated in Z_o , there will be perfect matching and the waves are completely absorbed without causing any reflections. Therefore, the reflected waves b_2 into the junction from 2nd line, b_3 into the junction from 3rd line and so on upto b_n into the junction from nth line will all be equal to zero.

$$\text{i.e., } b_2 = b_3 = \dots = b_n = 0.$$

If b_i represents the wave going back to the source, then the contribution to b_i will be only from the reflected wave b_1 from the 1st line, due to mismatch since the first line is terminated in an impedance $Z_L \neq Z_o$.

$$\therefore b_i = b_1 = (\text{reflection coefficient}) (a_1)$$

$$\therefore b_i = (S_{i1}) (a_1) \quad \dots (4.16)$$

Where S_{i1} = reflection coefficient of the 1st line due to the source connected at the i^{th} line. [The first subscript "i" denotes the source connected at i^{th} line and the second subscript "1", the reflection from the 1st line].

When all the $(n - 1)$ number of lines are terminated in an impedance other than Z_o , then there will be reflections from all the lines. Then the total contribution to the outward travelling wave in the i^{th} line due to mismatch at all the lines, is given by

$$b_i = S_{i1} a_1 + S_{i2} a_2 + S_{i3} a_3 + \dots + S_{in} a_n$$

Since " i " can be any line from 1 to n , by allowing i to vary we get

$$\left. \begin{aligned} b_1 &= S_{11} a_1 + S_{12} a_2 + S_{13} a_3 + \dots + S_{1n} a_n \\ b_2 &= S_{21} a_1 + S_{22} a_2 + S_{23} a_3 + \dots + S_{2n} a_n \\ b_3 &= S_{31} a_1 + S_{32} a_2 + S_{33} a_3 + \dots + S_{3n} a_n \\ b_n &= S_{n1} a_1 + S_{n2} a_2 + S_{n3} a_3 + \dots + S_{nn} a_n \end{aligned} \right\} \dots (4.17)$$

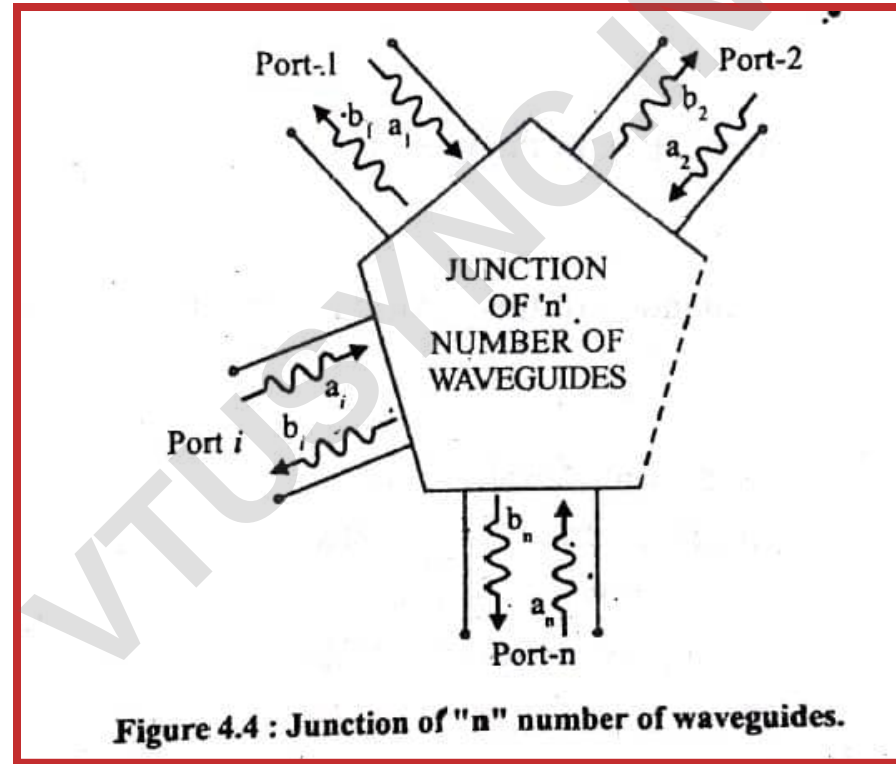
These set of equations (4.17) can be represented in the matrix form as

$$\begin{bmatrix} b_1 \\ b_2 \\ b_3 \\ \vdots \\ b_n \end{bmatrix} = \begin{bmatrix} S_{11} & S_{12} & S_{13} & \dots & S_{1n} \\ S_{21} & S_{22} & S_{23} & \dots & S_{2n} \\ S_{31} & S_{32} & S_{33} & \dots & S_{3n} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ S_{n1} & S_{n2} & S_{n3} & \dots & S_{nn} \end{bmatrix} \begin{bmatrix} a_1 \\ a_2 \\ a_3 \\ \vdots \\ a_n \end{bmatrix} \quad \dots (4.18)$$

or $[b] = [S] [a] \quad \dots (4.19)$

Where $[b]$ represents the column matrix corresponding to reflected waves. $[a]$ represents the column matrix corresponding to incident waves and $[S]$ represents the "Scattering matrix" corresponding to reflection coefficients which is a " $n \times n$ " square matrix.

S-MATRIX REPRESENTATION OF MULTIPORT NETWORK



S-MATRIX REPRESENTATION OF MULTIPORT NETWORK

Let us now consider a junction of "n" number of rectangular waveguides as shown in figure 4.4. In this case, all "a's" represent the incident waves at respective ports and all "b's" the reflected waves from the microwave junction coming out of the respective ports. In this case also, equations (4.18) and (4.19) are still valid where S_{ij} and S_{ii} have the following meanings:

S_{ij} = Scattering coefficient corresponding to the input power applied at the i^{th} port and output power coming out of j^{th} port and

S_{ii} = Scattering coefficient corresponding to the power applied at the i^{th} port and output taken out of i^{th} port itself. This coefficient is a measure of amount of mismatch between the i^{th} port and the junction.

As an example, let us consider a two-port network as shown in figure 4.5.

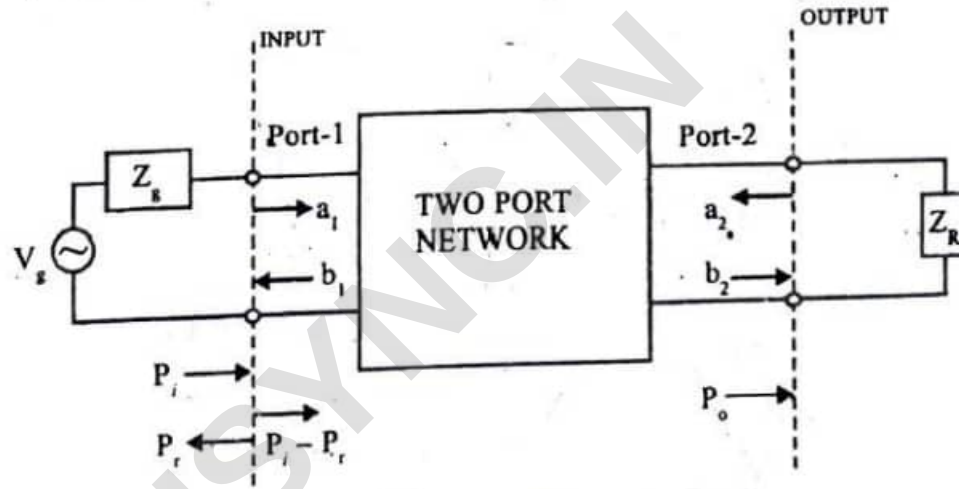


Figure 4.5 : Illustrating two-port network

The relationship between the incident and reflected waves in terms of scattering coefficients can be written as

$$b_1 = S_{11} a_1 + S_{12} a_2 \quad \dots (4.20)$$

$$b_2 = S_{21} a_1 + S_{22} a_2 \quad \dots (4.21)$$

From these equations, the scattering coefficients are found as

$$\begin{aligned}b_1 &= S_{11} a_1 + S_{12} a_2 \\b_2 &= S_{21} a_1 + S_{22} a_2\end{aligned}$$

From these equations, the scattering coefficients are found as

$$S_{11} = \left. \frac{b_1}{a_1} \right|_{a_2=0} = \text{reflection coefficient at port-1 when port-2 is terminated with a matched load } (a_2 = 0)$$

$$S_{22} = \left. \frac{b_2}{a_2} \right|_{a_1=0} = \text{reflection coefficient at port-2 when port-1 is terminated with a matched load } (a_1 = 0)$$

$$S_{12} = \left. \frac{b_1}{a_2} \right|_{a_1=0} = \text{attenuation of the wave travelling from port-2 to port-1.}$$

$$S_{21} = \left. \frac{b_2}{a_1} \right|_{a_2=0} = \text{attenuation of the wave travelling from port-1 to port-2.}$$

As an example, let us consider a two-port network as shown in figure 4.5.

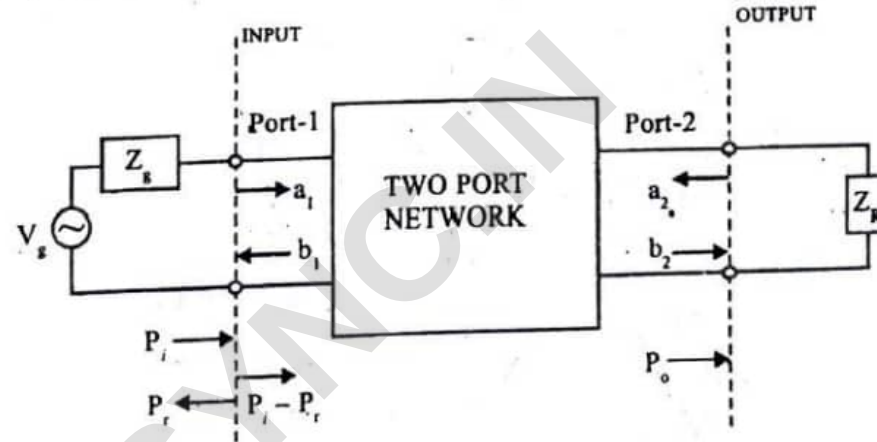


Figure 4.5 : Illustrating two-port network

In figure 4.5, we have

P_i = incident power at port-1

P_r = power reflected by the network coming out of port-1 itself.

P_o = output power at port-2.

The various losses can be expressed in terms of S-parameters as given below:

$$1. \text{ Insertion loss in dB} = 10 \log_{10} \frac{P_i}{P_o} \quad \dots (4.22)$$

But

$$P_i \propto |a_1|^2$$

$$P_o \propto |b_2|^2$$

$$\therefore \frac{P_i}{P_o} = \frac{|a_1|^2}{|b_2|^2} = \frac{1}{\left| \frac{b_2}{a_1} \right|^2} = \frac{1}{|S_{21}|^2} = \frac{1}{|S_{12}|^2} \quad \dots (4.23)$$

[NOTE : From symmetric property $|S_{12}| = |S_{21}|$ which will be discussed later].

Substituting equation (4.23) in (4.22), we get

$$\begin{aligned} \text{Insertion loss in dB} &= 10 \log_{10} \frac{1}{|S_{21}|^2} \\ &= 20 \log_{10} \frac{1}{|S_{21}|} = 20 \log_{10} \frac{1}{|S_{12}|} \quad \dots (4.24) \end{aligned}$$

2. Transmission loss (or attenuation loss) in dB

$$= 10 \log_{10} \frac{P_i - P_r}{P_o}$$

$$= 10 \log_{10} \frac{|a_1|^2 - |b_1|^2}{|b_2|^2}$$

$$= 10 \log_{10} \frac{\frac{|a_1|^2 - |b_1|^2}{|a_1|^2}}{\frac{|b_2|^2}{|a_1|^2}} = 10 \log_{10} \frac{1 - \frac{|b_1|^2}{|a_1|^2}}{\frac{|b_2|^2}{|a_1|^2}}$$

$$= 10 \log_{10} \frac{1 - |S_{11}|^2}{|S_{21}|^2}$$

..... (4.25)

3. Reflection loss in dB = $10 \log_{10} \frac{P_i}{P_i - P_r}$

$$= 10 \log_{10} \frac{|a_1|^2}{|a_1|^2 - |b_1|^2}$$

$$= 10 \log_{10} \frac{1}{1 - \frac{|b_1|^2}{|a_1|^2}}$$

$$= 10 \log_{10} \frac{1}{1 - |S_{11}|^2}$$

..... (4.26)

4. Return loss in dB = $10 \log_{10} \frac{P_i}{P_r}$

$$= 10 \log_{10} \frac{|a_1|^2}{|b_1|^2}$$
$$= 10 \log_{10} \frac{1}{|S_{11}|^2}$$
$$= 20 \log_{10} \frac{1}{|S_{11}|} \quad \dots (4.27)$$

Properties of S Matrix:

Seventh Semester B.E. Degree Examination, Dec.2018/Jan.2019 Microwaves and Antennas

Time: 3 hrs.

Max. Marks: 80

Note: Answer any FIVE full questions, choosing
ONE full question from each module.

Module-1

- 1 a. Derive the general transmission line equation to find voltage and current on the line in terms of position 'z' and time 't'. (07 Marks)
- b. Describe the different mode curve in the case of reflex klystron. (05 Marks)
- c. A transmission line has a characteristic impedance of $50 + j0.01\Omega$ and terminated in a load impedance of $73 - j42.5\Omega$ calculate : i) reflection coefficient ii) SWR. (04 Marks)

OR

- 2 a. Define reflection coefficient. Derive the equation for reflection coefficient at the load end at a distance 'd' from the load. (06 Marks)
- b. Describe the mechanism of oscillation of reflex klystron. (06 Marks)
- c. A transmission line has the following parameters : $R = 2\Omega/m$, $G = 0.5\text{mmho}/m$, $f = 1\text{GHz}$, $L = 8\text{nH}/m$, $C = 0.23\text{pF}/m$. Calculate : i) characteristic impedance ii) propagation constant. (04 Marks)

Module-2

- 3 a. State and explain the properties of S - matrix. (07 Marks)
- b. With a neat diagram, explain the working of precession type variable attenuator. (06 Marks)

Seventh Semester B.E. Degree Examination, June/July 2019 Microwave and Antennas

Time: 3 hrs.

Max. Marks: 80

Note: Answer any FIVE full questions, choosing ONE full question from each module.

Module-1

1. a. Discuss mechanism of oscillation in Reflex Klystron with schematic. (06 Marks)
- b. A Reflex Klystron is to be operated at 10GHz with dc beam voltage 300V, repeller space 0.1cm for $1\frac{3}{4}$ mode. Calculate $P_{at,max}$ and corresponding repeller voltage for a beam current of 20mA. (05 Marks)
- c. A transmission line has the following parameters $R = 2\Omega/m$, $G = 0.5mho/m$, $f = 1GHz$, $L = 8nH/m$ and $C = 0.23PF$. Calculate its characteristic impedance and propagation constant. (05 Marks)

OR

2. a. A line of 400Ω is connected to a load of $200 + j300\Omega$ which is excited by a matched generator at 800MHz. Find the location and length of a single stub nearest to the load to produce an impedance match. (08 Marks)
- b. A certain transmission line has a characteristic impedance of $75 + j0.01\Omega$ and is terminated in a load impedance of $75 + j50\Omega$. Compute: i) Reflection coefficient ii) The transmission coefficient. (04 Marks)
- c. What are the high frequency limitations of conventional vacuum tube / transistors? (04 Marks)

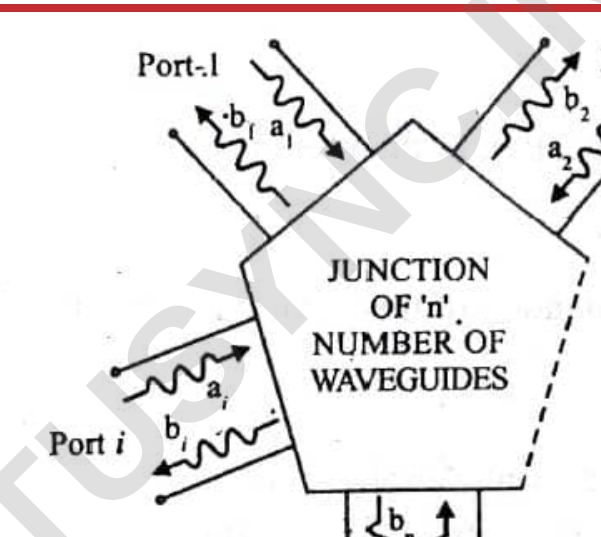
Module-2

3. a. Show that impedance and admittance matrices are symmetrical for a reciprocal junction. (06 Marks)
- b. In a H-plane T-junction, compute power delivered to the loads 40ohm and 60ohm connected to arms 1 and 2 when 10mw power is delivered to matched port 3. Assume characteristic impedance of line = 50ohm. (04 Marks)
- c. Two transmission lines of characteristic impedance Z_1 and Z_2 are joined at plane pp' . Express S-parameters in terms of impedances. (06 Marks)

OR

4. a. Discuss the following properties of S-parameters:
 - i) Symmetry of [S] for a reciprocal network
 - ii) Unitary property for a lossless junction.

Thus $S_{ii} = 0$. If all the ports are perfectly matched, all will be zero.



The diagram shows a central pentagonal block labeled "JUNCTION OF 'n' NUMBER OF WAVEGUIDES". Four ports are connected to this junction:

- Port-1** (top-left): Two waveguide lines. The incident wave is labeled a_1 (arrow pointing towards the junction) and the reflected wave is labeled b_1 (arrow pointing away from the junction).
- Port-2** (top-right): Two waveguide lines. The incident wave is labeled a_2 (arrow pointing towards the junction) and the reflected wave is labeled b_2 (arrow pointing away from the junction).
- Port i** (bottom-left): Two waveguide lines. The incident wave is labeled a_i (arrow pointing towards the junction) and the reflected wave is labeled b_i (arrow pointing away from the junction).
- Bottom Port**: Two vertical waveguide lines. The incident wave is labeled a_n (arrow pointing upwards towards the junction) and the reflected wave is labeled b_n (arrow pointing downwards away from the junction).

Figure 4.4 : Junction of "n" number of waveguides.

Property (2) : Symmetric Property of S-matrix:- If a microwave junction satisfies reciprocity condition and if there are no active devices, then S parameters are equal to their corresponding transposes.

$$\text{i.e., } S_{ij} = S_{ji} \quad \text{..... (4.28)}$$

Proof : The steady-state voltage and current at the i^{th} port are given by

$$V_i = V_i^+ + V_i^- \quad \text{..... (4.29)}$$

and

$$I_i = \frac{V_i^+}{Z_{oi}} - \frac{V_i^-}{Z_{oi}} \quad \text{..... (4.30)}$$

where

V_i^+ = incident voltage at the i^{th} port

V_i^- = reflected voltage from the i^{th} port.

Z_{oi} = characteristic impedance

From equations (4.29) and (4.30), V_i^+ and V_i^- can be found to be

$$V_i^+ = \frac{1}{2} (V_i + Z_{oi} I_i) \quad \text{..... (4.31)}$$

$$V_i^- = \frac{1}{2} (V_i - Z_{oi} I_i) \quad \text{..... (4.32)}$$

The average incident power (complex) at the i^{th} port is

$$\frac{1}{2} V_i I_i^* = \frac{|V_i^+|^2}{2 Z_{oi}^*} \quad \dots (4.33)$$

The normalized incident and reflected voltages at the i^{th} port can be defined as

$$a_i = \frac{V_i^+}{\sqrt{Z_{oi}}} = \frac{1}{2} \left(\frac{V_i}{\sqrt{Z_{oi}}} + \sqrt{Z_{oi}} I_i \right) \quad \dots (4.34)$$

$$b_i = \frac{V_i^-}{\sqrt{Z_{oi}}} = \frac{1}{2} \left(\frac{V_i}{\sqrt{Z_{oi}}} - \sqrt{Z_{oi}} I_i \right) \quad \dots (4.35)$$

If the characteristic impedance is also normalized so that $\sqrt{Z_{oi}} = 1$, then

$$a_i = \frac{1}{2} (V_i + I_i) \quad \dots (4.36)$$

and

$$b_i = \frac{1}{2} (V_i - I_i) \quad \dots (4.37)$$

Adding (4.36) and (4.37), $V_i = a_i + b_i$ (4.38)

Subtracting (4.37) from (4.36), $I_i = a_i - b_i$ (4.39)

The n-port voltages are linearly related to the n-port currents by the "**impedance matrix** Z_{ij} " of the junction given by

$$V_i = \sum_{j=1}^n Z_{ij} I_j \text{ for } i = 1, 2, 3, \dots, n \quad \text{..... (4.40)}$$

Equation (4.40) can also be written in the matrix form as

$$\begin{bmatrix} V_1 \\ V_2 \\ \vdots \\ V_n \end{bmatrix} = \begin{bmatrix} Z_{11} & Z_{12} & \dots & Z_{1n} \\ Z_{21} & Z_{22} & \dots & Z_{2n} \\ \vdots & \vdots & & \vdots \\ Z_{n1} & Z_{n2} & \dots & Z_{nn} \end{bmatrix} \begin{bmatrix} I_1 \\ I_2 \\ \vdots \\ I_n \end{bmatrix} \quad \text{..... (4.41)}$$

$$\text{Symbolically } [V] = [Z] [I] \quad \text{..... (4.42)}$$

Using equation (4.40) in (4.36), we get

$$\begin{aligned}
 a_i &= \frac{1}{2} \left[\sum_{j=1}^n Z_{ij} I_j + I_i \right] \\
 &= \frac{1}{2} \left[\sum_{j=1}^n Z_{ij} I_j + \sum_{j=1}^n \delta_{ij} I_j \right] \\
 a_i &= \frac{1}{2} \sum_{j=1}^n (Z_{ij} + \delta_{ij}) I_j
 \end{aligned}
 \tag{4.43}$$

Similarly, by using equation (4.40) in (4.37), we get

$$b_i = \frac{1}{2} \sum_{j=1}^n (Z_{ij} - \delta_{ij}) I_j
 \tag{4.44}$$

Where δ_{ij} is known as "**Kronecker delta**" given by

$$\delta_{ij} = 1 \text{ if } i = j
 \tag{4.45}$$

and

$$\delta_{ij} = 0 \text{ if } i \neq j
 \tag{4.46}$$

In matrix notation, equations (4.43) and (4.44) can be written as

$$[a] = \frac{1}{2} \{[Z] + [U]\} [I] \quad \dots (4.47)$$

and

$$[b] = \frac{1}{2} \{[Z] - [U]\} [I] \quad \dots (4.48)$$

where

$[U]$ = identity matrix

From equation (4.47),

$$\{[Z] + [U]\} [I] = 2 [a]$$

Pre multiplying both sides by $\{[Z] + [U]\}^{-1}$, we get

$$\{[Z] + [U]\}^{-1} \{[Z] + [U]\} [I] = 2 \{[Z] + [U]\}^{-1} [a]$$

$$\therefore [I] = 2 \{[Z] + [U]\}^{-1} [a] \quad \dots (4.49)$$

Using equation (4.49) in (4.48), we get

$$[b] = \{[Z] - [U]\} \{[Z] + [U]\}^{-1} [a] \quad \text{..... (4.50)}$$

From equation (4.19), we have

$$[b] = [S] [a] \quad \text{..... (4.51)}$$

Comparing equations (4.50) and (4.51), we get

$$[S] = \{[Z] - [U]\} \{[Z] + [U]\}^{-1} \quad \text{..... (4.52)}$$

$$\text{Let } [Z] - [U] = P \quad \text{..... (4.53)}$$

$$\text{and } [Z] + [U] = Q \quad \text{..... (4.54)}$$

Using equations (4.53) and (4.54) in (4.52), we get

$$[S] = [P] [Q]^{-1} \quad \text{..... (4.55)}$$

For a reciprocal network, the Z-matrix is symmetric. Hence [P] and [Q] are also symmetric so that

$$[P][Q] = [Q][P] \quad \dots (4.56)$$

Pre-and post multiplying both sides of the above equation by $[Q]^{-1}$, we get

$$[Q]^{-1}[P][Q][Q]^{-1} = [Q]^{-1}[Q][P][Q]^{-1}$$

$$\therefore [Q]^{-1}[P] = [P][Q]^{-1} = [S] \quad \dots (4.57)$$

Consider, now the transpose of [S] in equation (4.52) to get

$$[S]_T = \{[Z] + [U]\}_T^{-1} \cdot \{[Z] - [U]\}_T \quad [\text{Since } [AB]_T = [B]_T [A]_T] \quad \dots (4.58)$$

Since [Z] matrix is symmetrical

$$\{[Z] + [U]\}_T^{-1} = \{[Z] + [U]\}^{-1} \quad \dots (4.59)$$

$$\text{and } \{[Z] - [U]\}_T = \{[Z] - [U]\} \quad \dots (4.60)$$

Using equations (4.59) and (4.60) in (4.58), we get

$$[S]_T = \{[Z] + [U]\}^{-1} \{[Z] - [U]\}$$

$$[S]_T = [Q]^{-1}[P] \quad \dots (4.61)$$

Comparing equations (4.57) and (4.61), we can conclude that

$$[S] = [S]_T \quad \dots (4.62)$$

This means that the terms S_{ij} and S_{ji} of the [S] matrix are equal and thus the [S] matrix is a symmetric matrix.

Property (3):- Unitary property for a lossless junction:-

This property states that for any lossless network, the sum of the products of each term of any one row or any one column of the [S] matrix with its complex conjugate is unity.

Proof:- From the principle of conservation of energy, if the junction is lossless, then the power input must be equal to power output. The incident and reflected waves are related to the incident and reflected voltages by

$$a = \frac{V^+}{\sqrt{Z_0}} \quad \text{..... (4.63)}$$

and
$$b = \frac{V^-}{\sqrt{Z_0}} \quad \text{..... (4.64)}$$

$$\therefore \text{The incident power} = P^+ = \frac{1}{2} a a^* = \frac{1}{2} |a|^2 \quad \text{..... (4.65)}$$

$$\text{and Reflected power} = P^- = \frac{1}{2} b b^* = \frac{1}{2} |b|^2 \quad \text{..... (4.66)}$$

When the junction is lossless, then no real power can be delivered to the network. Thus, if the characteristic impedances of all the ports are identical and assumed to be unity (perfectly normalized), the average power delivered to junction is **zero**.

$$\therefore P_{av} = \frac{1}{2} \operatorname{Re} \sum_{i=1}^n V_i I_i^* = 0 \quad \text{..... (4.67)}$$

where I_i^* = complex conjugate of I_i

Using equations (4.38) and (4.39) in (4.67), we get

$$\begin{aligned} \frac{1}{2} \operatorname{Re} \sum_{i=1}^n (a_i + b_i) (a_i - b_i)^* &= 0 \\ \therefore \frac{1}{2} \operatorname{Re} \sum_{i=1}^n (a_i + b_i) (a_i^* - b_i^*) &= 0 \\ \therefore \frac{1}{2} \operatorname{Re} \sum_{i=1}^n [(a_i a_i^* - b_i b_i^*) + (a_i^* b_i - a_i b_i^*)] &= 0 \\ \therefore \sum_{i=1}^n [(a_i a_i^* - b_i b_i^*)] &= 0 \quad \text{..... (4.68)} \end{aligned}$$

Equation (4.68) can be written in matrix form as

$$[a] [a]^* - [b] [b]^* = [0] \quad \text{..... (4.69)}$$

From equation (4.19), we have

$$[b] = [S] [a] \quad \text{..... (4.70)}$$

Taking complex conjugate on both sides

$$[b]^* = [S]^* [a]^* \quad \text{..... (4.71)}$$

Using equations (4.70) and (4.71) in (4.69), we get

$$[a] [a]^* - [S] [a] [S]^* [a]^* = 0$$

$$[a] [a]^* \{ [U] - [S] [S]^* \} = 0 \quad \text{..... (4.72)}$$

Since $[a] [a]^*$ cannot be zero, we must have

$$[U] - [S] [S]^* = 0$$

$$\therefore [S] [S]^* = [U] \quad \text{..... (4.73)}$$

Where $[U]$ is the unit matrix or identity matrix. Pre-multiplying both sides of equation (4.73) by $[S]^{-1}$, we get

$$[S]^{-1} [S] [S]^* = [S]^{-1} [U]$$

$$\therefore [S]^* = [S]^{-1} \quad \text{..... (4.74)}$$

Any matrix satisfying equation (4.74) is said to be "**UNITARY**". In other words, a matrix is said to satisfy "**unitary property**" when its complex conjugate is equal to its inverse.

From equation (4.62),

$$[S] = [S]_T$$

Using this in equation (4.74), we get

$$[S]^* = \{[S]_T\}^{-1} \quad \text{..... (4.75)}$$

A matrix that satisfies the condition of (4.75) is also termed as "**unitary matrix**".

The matrix equation of (4.75) can be written in summation form as

$$\sum_{k=1}^n S_{ki} S_{kj}^* = \delta_{ij} \text{ for all } i, j \quad \dots (4.76)$$

Where $\delta_{ij} = 1$ if $i = j$

and $\delta_{ij} = 0$ if $i \neq j$

(a) when $i = j$, equation (4.76) reduces to

$$\sum_{k=1}^n S_{ki} S_{ki}^* = 1 \quad \dots (4.77)$$

This property of equation (4.77) of [S] matrix is sometimes called "**UNITY PROPERTY**".

(b) When $i \neq j$, equation (4.76) reduces to

$$\sum_{k=1}^n S_{ki} S_{kj}^* = 0 \quad \dots (4.78)$$

This property of equation (4.78) of [S] matrix is sometimes called "**ZERO PROPERTY**".

In words, equation (4.77) states that the product of any column of [S] with the complex conjugate of that column gives unity, while equation (4.78) states that the product of any column of [S] with the complex conjugate of a different column gives zero.

Property (4) :- PHASE-SHIFT PROPERTY

Complex S-parameters of a network are defined with respect to the positions of the port or reference planes. For a two-port network with unprimed reference planes 1 and 2 as shown in figure 4.6, the S-parameters have definite complex values

$$[S] = \begin{bmatrix} S_{11} & S_{12} \\ S_{21} & S_{22} \end{bmatrix} \quad \dots (4.79)$$

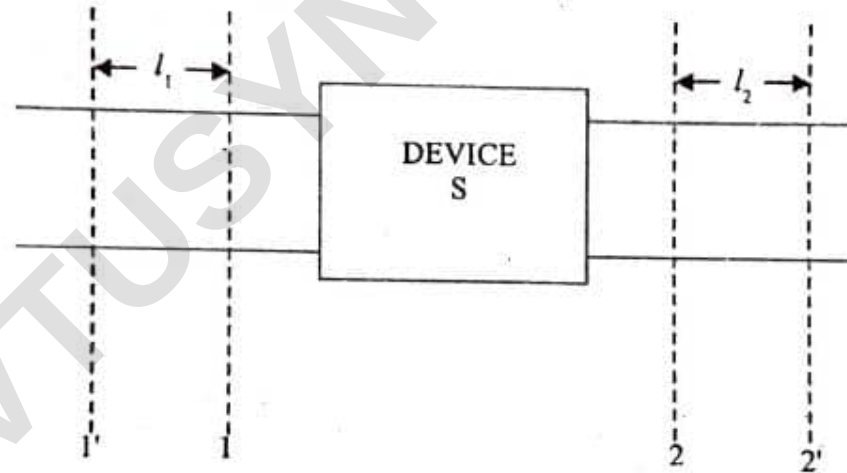


Figure 4.6 : Illustrating phase shift property of S

When the reference planes 1 and 2 are shifted outward to 1' and 2' by electrical phase shifts $\phi_1 = \beta_1 l_1$ and $\phi_2 = \beta_2 l_2$ respectively, then the new wave variables are $a_1, e^{j\phi_1}, b_1 e^{-j\phi_1}$ and $a_2 e^{j\phi_2}$ and $b_2 e^{-j\phi_2}$. The new S-matrix S' is then given by

$$[S'] = \begin{bmatrix} e^{-j\phi_1} & 0 \\ 0 & e^{-j\phi_2} \end{bmatrix} [S] \begin{bmatrix} e^{-j\phi_1} & 0 \\ 0 & e^{-j\phi_2} \end{bmatrix} \quad \dots (4.80)$$

This property is valid for any number of ports.

$\therefore$ For "n" number of ports,

$$[S'] = \begin{bmatrix} e^{-j\phi_1} & 0 & \dots & 0 \\ 0 & e^{-j\phi_2} & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ \vdots & \vdots & \vdots & \vdots \\ 0 & 0 & \dots & e^{-j\phi_n} \end{bmatrix} [S] \begin{bmatrix} e^{-j\phi_1} & 0 & \dots & 0 \\ 0 & e^{-j\phi_2} & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ \vdots & \vdots & \vdots & \vdots \\ 0 & 0 & \dots & e^{-j\phi_n} \end{bmatrix} \quad \dots (4.81)$$

The above property is called the **"PHASE SHIFT PROPERTY"** applicable to a shift of reference planes.

Microwave Passive Devices

- Introduction
- Coaxial Connectors and Adapters
- Attenuators
- Phase Shifters
- Waveguide Tees
- Magic tees

Suggested Reading

Text I: “Microwave Engineering” -
Annapurna Das, Sisir K Das, TMH Publication, 2nd Edition,
2010,

Text I:

Sections: 6.4.2, 6.4.14, 6.4.15, 6.4.16

CO-AXIAL CABLES, CONNECTORS AND ADAPTERS

CO-AXIAL CABLES, CONNECTORS AND ADAPTERS

Coaxial Cables

Microwave components and devices are interconnected using these co-axial cables of suitable length and operated at microwave frequencies. The characteristics and operation of co-axial line is already described in chapter-1. In this section let us consider some practical aspects of these co-axial cables. TEM mode is propagated through the co-axial line and the outer conductor guides these signals in the dielectric space between itself and inner conductor. The outer conductor also acts as a shield to prevent the external signals to interfere with the internal signal. It also prevents the internal signal leakage. The co-axial cables usually possess characteristic impedance of either $50\ \Omega$ or $75\ \Omega$. Based on the structure of shielding, coaxial cables are classified into three basic types.

(i) **Flexible co-axial Cable:** Figure 5.1 shows the structure of flexible-type of co-axial cable consisting of low loss solid or foam type polyethylene dielectric. Electromagnetic shielding is provided for outer single braid or double braid of the flexible cable as shown, by using knitted metal wire mesh. The centre conductor usually consists of multi strand wire.

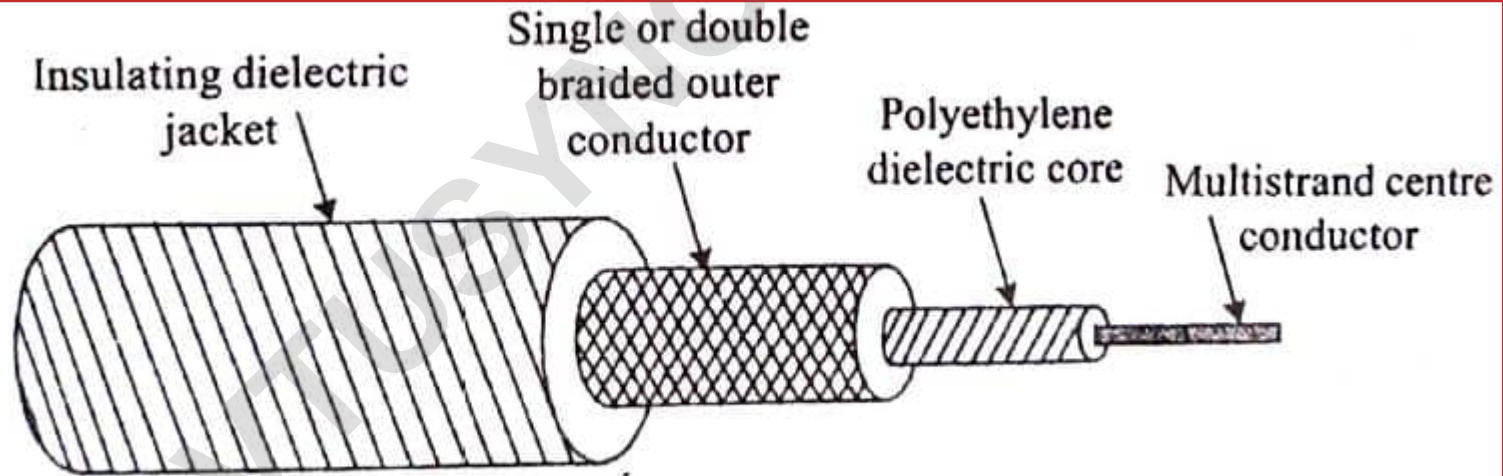


Figure 5.1 : Structure of flexible co-axial cable

(ii) **Semi-rigid co-axial cable:** Figure 5.2 shows the cross-sectional view of semi-rigid co-axial cable. Semi rigid co-axial cables make use of thin outer conductor made of copper and a strong inner conductor also made of copper. The region between the inner and outer conductor contains a solid dielectric. These cables can be slightly bent for convenient routing and are not as flexible as the first type.

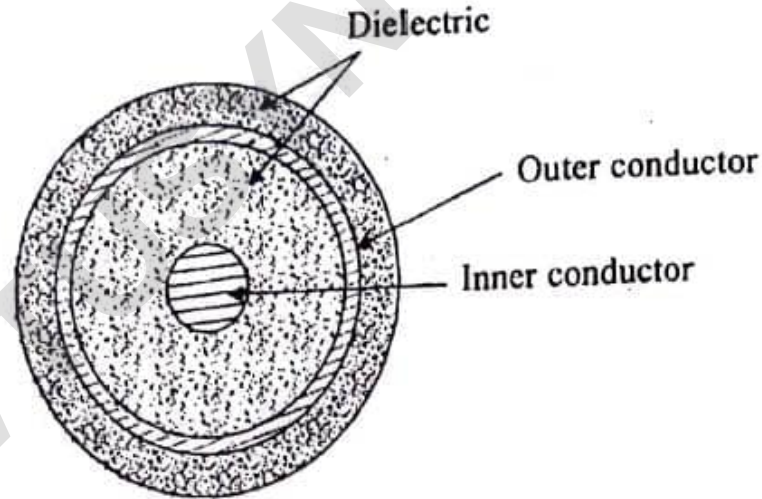
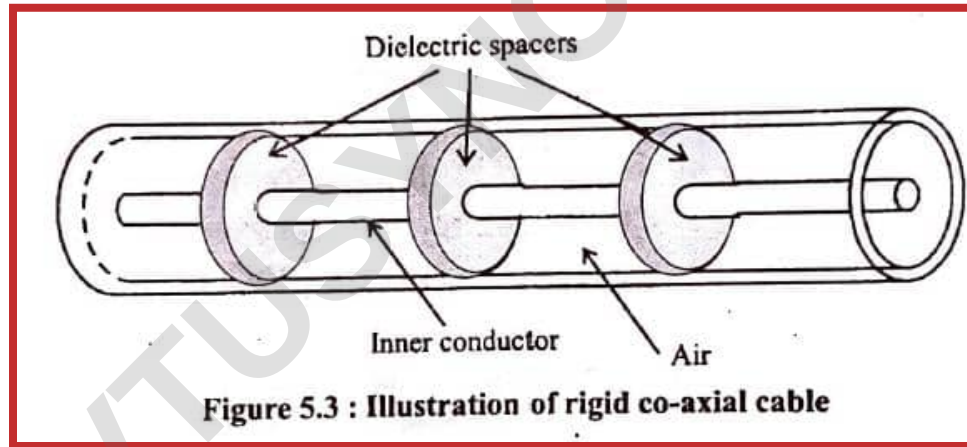


Figure 5.2 : Cross-section of semi-rigid co-axial cable

(iii) Rigid co-axial cable: Figure 5.3 shows the structure of a rigid co-axial cable consisting of inner and outer conductor with air as dielectric. To support the inner conductor at the centre dielectric spacers are introduced at regular intervals as shown. The thickness of these dielectric spacers is made small so that they do not produce significant discontinuities to the wave propagation.



Co-axial cables can be used upto microwave range of frequencies. Beyond these frequencies attenuation becomes very large (since attenuation increases with frequency) which makes co-axial cables unsuitable at higher frequencies.

Coaxial Connectors and Adaptors

Seventh Semester B.E. Degree Examination, Dec.2018/Jan.2019 Microwaves and Antennas

Time: 3 hrs.

Max. Marks: 80

Note: Answer any FIVE full questions, choosing
ONE full question from each module.

Module-1

1. a. Derive the general transmission line equation to find voltage and current on the line in terms of position 'z' and time 't'. (07 Marks)
- b. Describe the different mode curve in the case of reflex klystron. (05 Marks)
- c. A transmission line has a characteristic impedance of $50 + j0.01\Omega$ and terminated in a load impedance of $73 - j42.5\Omega$ calculate : i) reflection coefficient ii) SWR (04 Marks)

OR

2. a. Define reflection coefficient. Derive the equation for reflection coefficient at the load end at a distance 'd' from the load. (06 Marks)
- b. Describe the mechanism of oscillation of reflex klystron. (06 Marks)
- c. A transmission line has the following parameters : $R = 2\Omega/m$, $G = 0.5\text{mho/m}$, $f = 1\text{GHz}$, $L = 8\text{nH/m}$, $C = 0.23\text{pF/m}$. Calculate : i) characteristic impedance ii) propagation constant. (04 Marks)

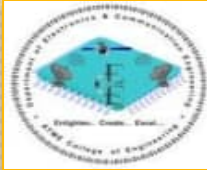
Module-2

3. a. State and explain the properties of S - matrix. (07 Marks)
- b. With a neat diagram, explain the working of precession type variable attenuator. (06 Marks)
- c. A 20mW signal is fed into one of the collinear port 1 of a lossless H-plane T junction. Calculate the power delivered through each port when other ports are terminated in matched load. (03 Marks)

OR

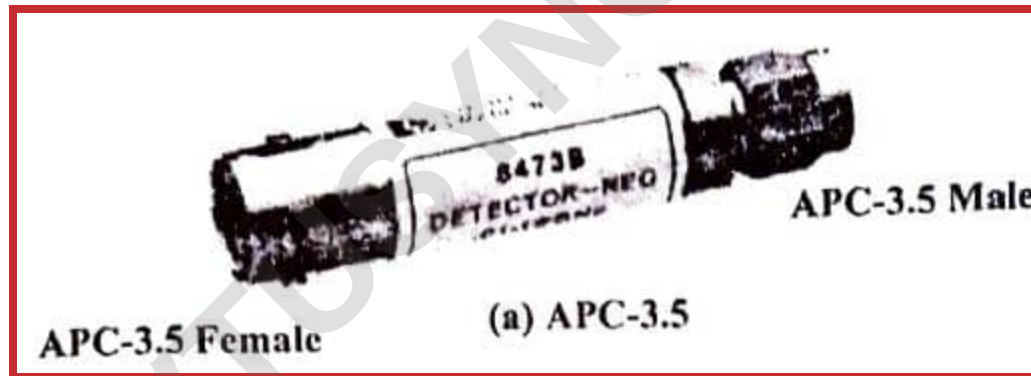
4. a. What is magic Tee? Derive its scattering matrix. (06 Marks)
- b. Discuss different types of coaxial connectors. (04 Marks)

Interconnection between co-axial cables and microwave components is achieved with the help of shielded standard connectors. The average circumference of the co-axial cable, for high frequency operation must be limited to about one wavelength.



1. APC 3.5 (Amphenol Precession Connector – 3.5 mm)
2. APC 7 - (Amphenol Precession Connector – 7 mm)
3. BNC (Bayonet Navy Connector)
4. SMA (Sub-Miniature A type)
5. SMC (Sub-Miniature C type)
6. TNC (Threaded Navy Connector)
7. Type N (Type – Navy) Connector

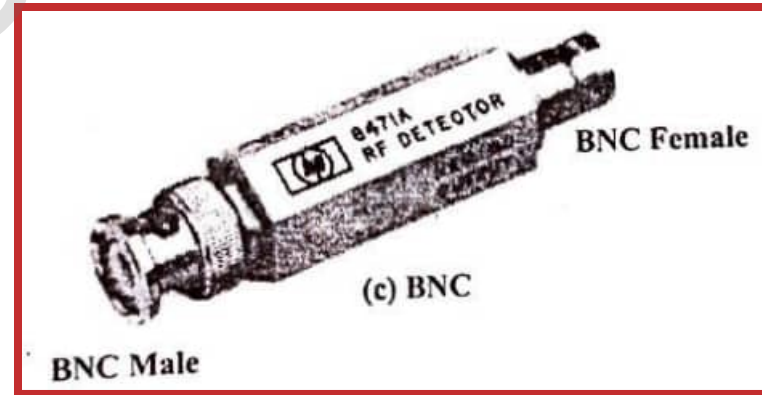
(a) APC 3.5 (Amphenol Precision Connector – 3.5 mm)



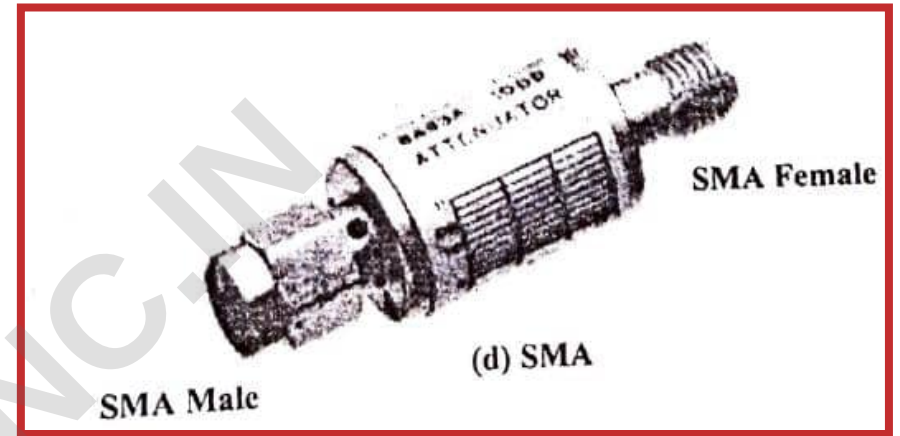
(b) APC – 7 (Amphenol Precision connector – 7 mm)



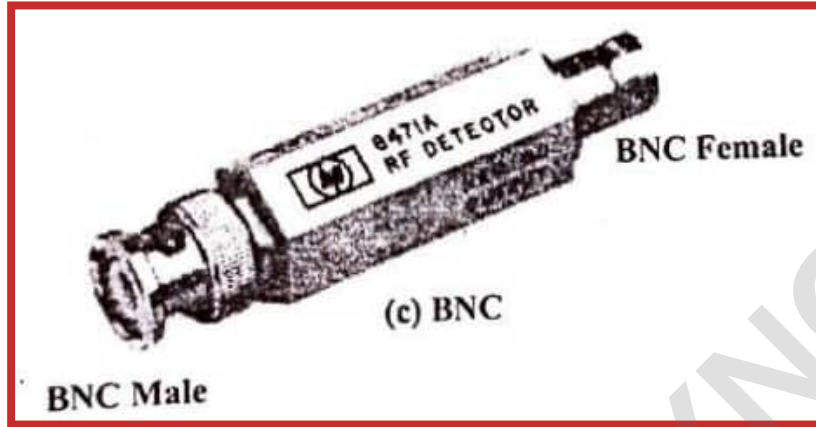
(c) BNC (Bayonet Navy Connector)



(d) SMA (Sub-Miniature A type)

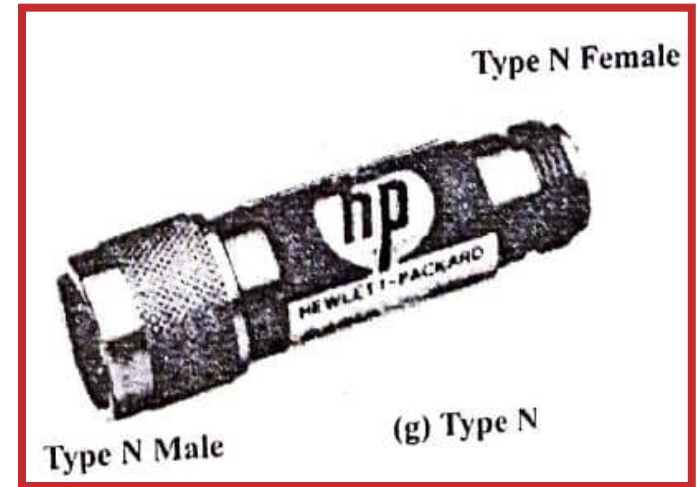


(e) SMC (Sub-Miniature C-type)



(f) TNC (Threaded Navy Connector)

(g) Type-N (Type-Navy) connector



(a) APC 3.5 (Amphenol Precision Connector – 3.5 mm)

HP (Hewlett - Packard) originally developed this connector, but it is now being manufactured by Amphenol. This connector can operate upto a frequency of 34 GHz and has a very low voltage standing wave ratio (VSWR). This connector provides repeatable connections and has $50\ \Omega$ characteristic impedance. The male or female of SMA connector (see below) can be connected to the opposite type of APC 3.5 connector.

(b) APC – 7 (Amphenol Precision connector – 7 mm)

This connector was also developed by HP but improved later by Amphenol. This connector provides repeatable connections and used for very accurate 50 ohm measurement applications. This connector provides a coupling mechanism without male or female distinction (i.e., sexless) and its VSWR is extremely low, less than 1.02 in the frequency range upto 18 GHz.

(c) BNC (Bayonet Navy Connector)

This connector was developed during World War II and used for military applications. It has characteristic impedance 50 to 75 Ω and is connected to flexible co-axial cable with diameters upto 0.635 cm. It is extensively used in almost all electronic measuring equipments upto 1 GHz of frequencies. BNC can be used even upto 4 GHz frequency and beyond that it starts radiating electromagnetic energy.

(d) SMA (Sub-Miniature A type)

This type of connector is also called OSM connector as it is manufactured by Omni-Spectra Inc. SMA connectors are used on components for microwave systems. The disadvantage with these connectors is that at high frequencies greater than 24 GHz, it introduces higher order modes and hence not used above 24 GHz.

(e) SMC (Sub-Miniature C-type)

This connector is manufactured by Sealectro Corporation and its size is smaller than SMA connector. It is a $50\ \Omega$ connector that connects flexible cables upto a diameter of 0.317 cm and used upto a frequency of 7 GHz.

(f) TNC (Threaded Navy Connector)

This connector is an improved version of BNC in the sense that it is threaded. This threading prevents radiation at high frequencies so that it can be used upto about 12 GHz frequency.

(g) Type-N (Type-Navy) connector

It is a $50\ \Omega$ or $75\ \Omega$ connector having a very low value of VSWR less than 1.02. This was developed during World War II and extensively used as a microwave measurement connector upto a frequency of 18 GHz.

ATTENUATORS

ATTENUATORS

In order to control power levels in a microwave system by partially absorbing the transmitted microwave signal, attenuators are employed. Resistive films (dielectric glass slab coated with aquadag) are used in the design of both fixed and variable attenuators.

There are two types of attenuators:

1. Fixed Attenuator
 2. Variable Attenuator
 - Movable vane variable attenuator
 - Flap attenuator
- ❖ Precision type variable attenuator

Aquadag is a trade name for a water-based colloidal graphite coating commonly used in cathode ray tubes (CRTs). It is manufactured by **Acheson Industries**, a subsidiary of **ICI**. The name is a shortened form of "Aqueous Deflocculated Acheson Graphite"^[1], but has become a generic term for conductive graphite coatings used in vacuum tubes. Other related products include Oildag, Electrodag and Molydag. Deflocculation refers to the distribution of powdered high purity graphite in an aqueous solution containing approximately 2% to 10% by weight of various Tannic/Gallotannic acid variants and separating the colloidal graphite suspension from the remaining unsuspended graphite particulates. The product names are often printed with DAG in upper case (e.g. AquaDAG). It is used as an electrically conductive coating on insulating surfaces, and as a lubricant.

Seventh Semester B.E. Degree Examination, Dec.2019/Jan.2020

Microwaves and Antennas

Time: 3 hrs.

Max. Marks: 80

Note: Answer any FIVE full questions, choosing ONE full question from each module.

Module-1

- 1 a. List four applications of Reflex Klystron. (04 Marks)
b. Derive transmission line equations in voltage and current forms. (06 Marks)
c. A transmission line is terminated in a resistive load of 1000Ω and has $L = 9\mu\text{H/m}$ and $C = 100\text{pF/m}$. Calculate reflection coefficient and standing wave ratio. (06 Marks)

OR

- 2 a. Define reflection coefficient. Derive an expression for reflection coefficient at load in terms of characteristic impedance and load impedance. (08 Marks)
b. Explain microwave system with the aid of a diagram. (08 Marks)

Module-2

- 3 a. For a two port network with mismatched load derive an expression for input reflection coefficient. (06 Marks)
b. Draw the diagram of Magic-Tee. Derive S-matrix of Magic Tee. (10 Marks)

OR

- 4 a. What is a reciprocal device? Write five point comparison among [S], [Z] and [Y] matrices. (06 Marks)
b. Given $[z] = \begin{bmatrix} 3 & 7 \\ 2 & 5 \end{bmatrix}$ Find S-matrix. (05 Marks)
c. Explain coaxial line fixed alternator with a diagram. (05 Marks)

Fixed Attenuator

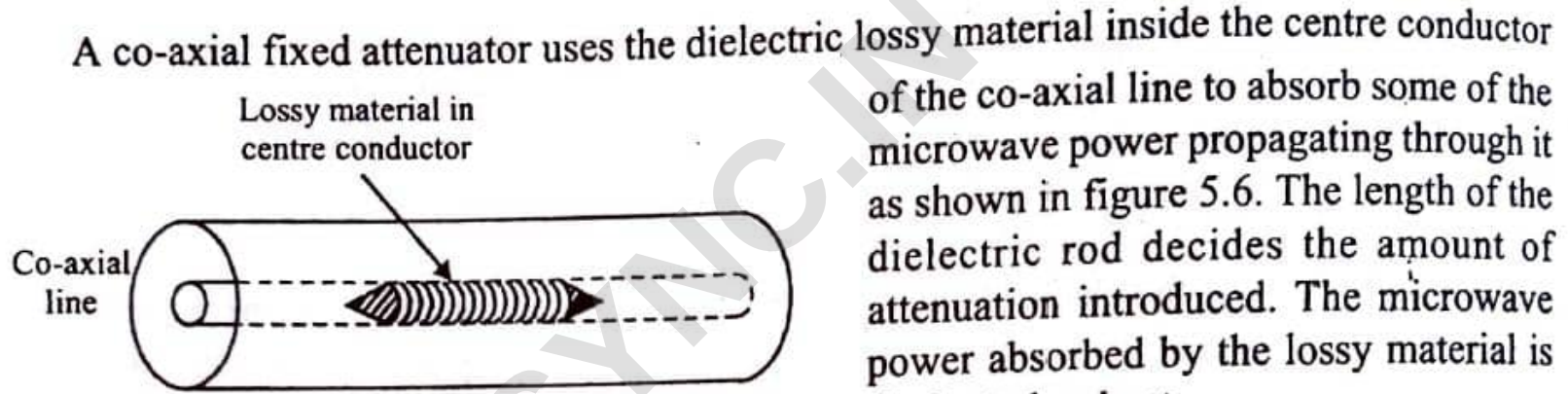
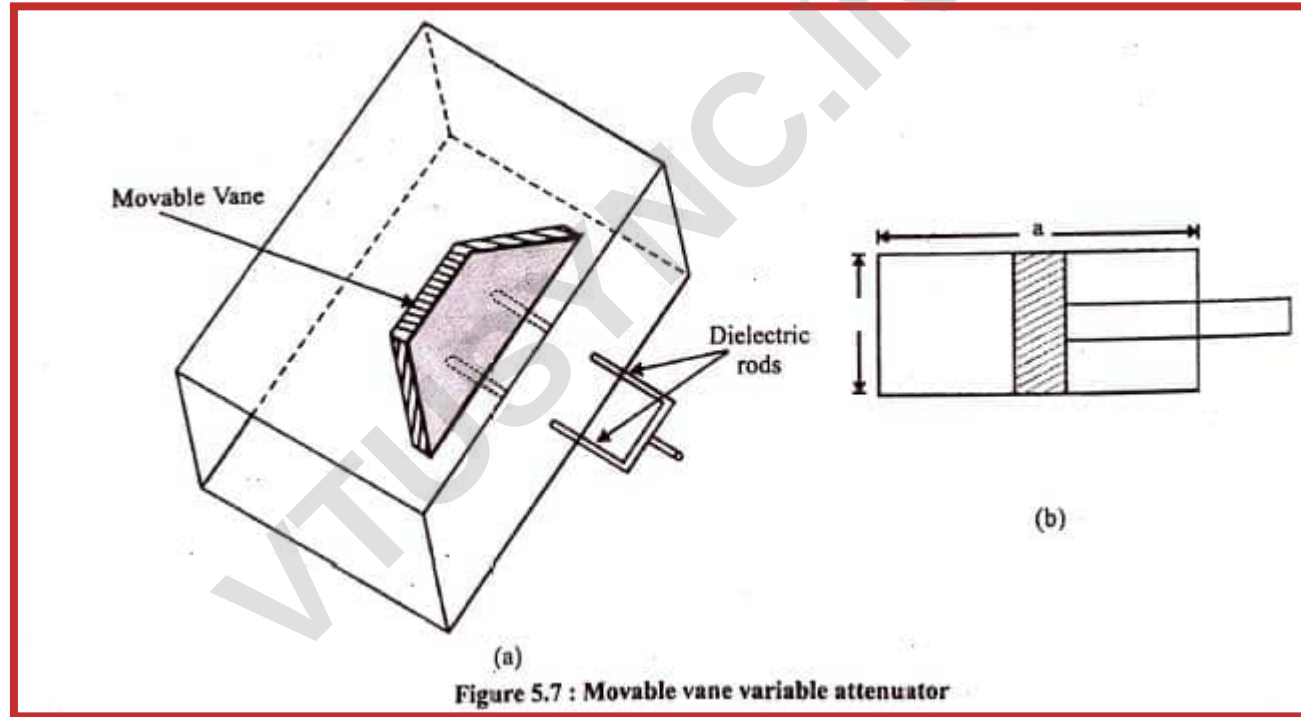


Figure 5.6 Co-axial line fixed attenuator

Variable Attenuator

- Movable vane variable attenuator



In waveguides, the dielectric slab coated with aduadag is placed at the centre of the waveguide parallel to the maximum E-field for dominant TE_{10} mode. Induced current on the lossy material due to incoming microwave signal, results in power dissipation, leading to attenuation of the signal. The dielectric slab is tapered at both ends upto a length of more than half wavelength to reduce reflections as shown in figure 5.7. The dielectric slab may be made movable along the breadth of the waveguide by supporting it with two dielectric rods separated by an odd multiple of quarter guide wavelength and perpendicular to electric field. When the slab is at the centre, then the attenuation is maximum (since the electric field is concentrated at the centre for TE_{10} mode) and when it is moved towards one side-wall, the attenuation goes on decreasing thereby controlling the microwave power coming out of the other port.

- Flap attenuator

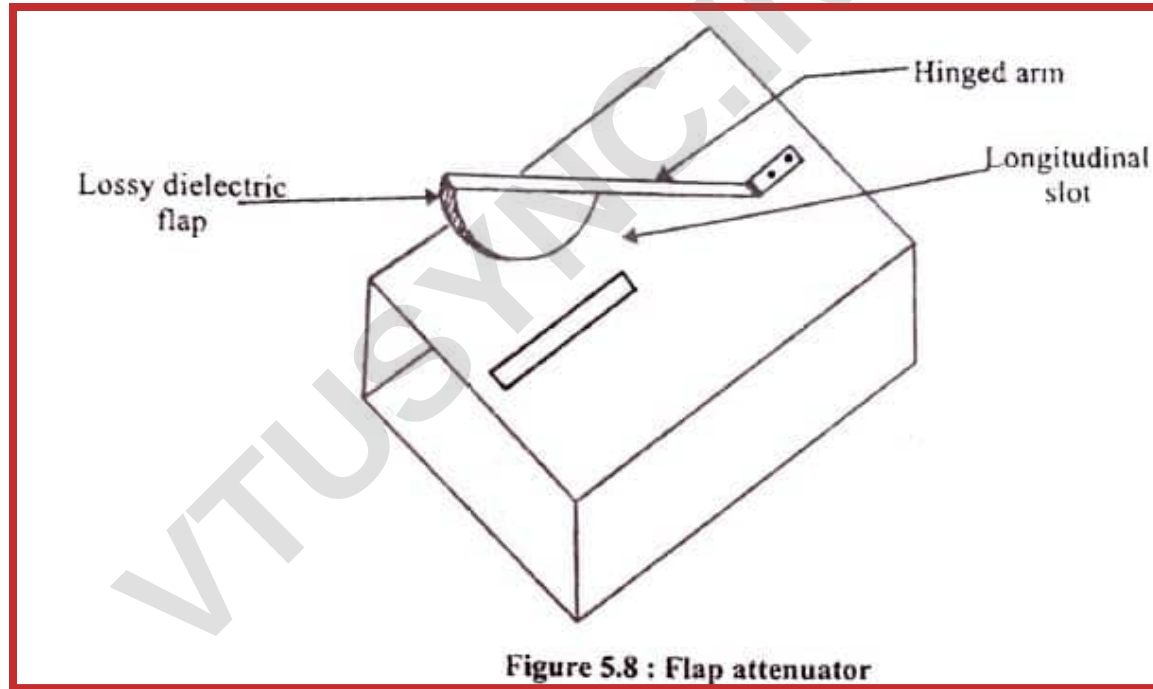
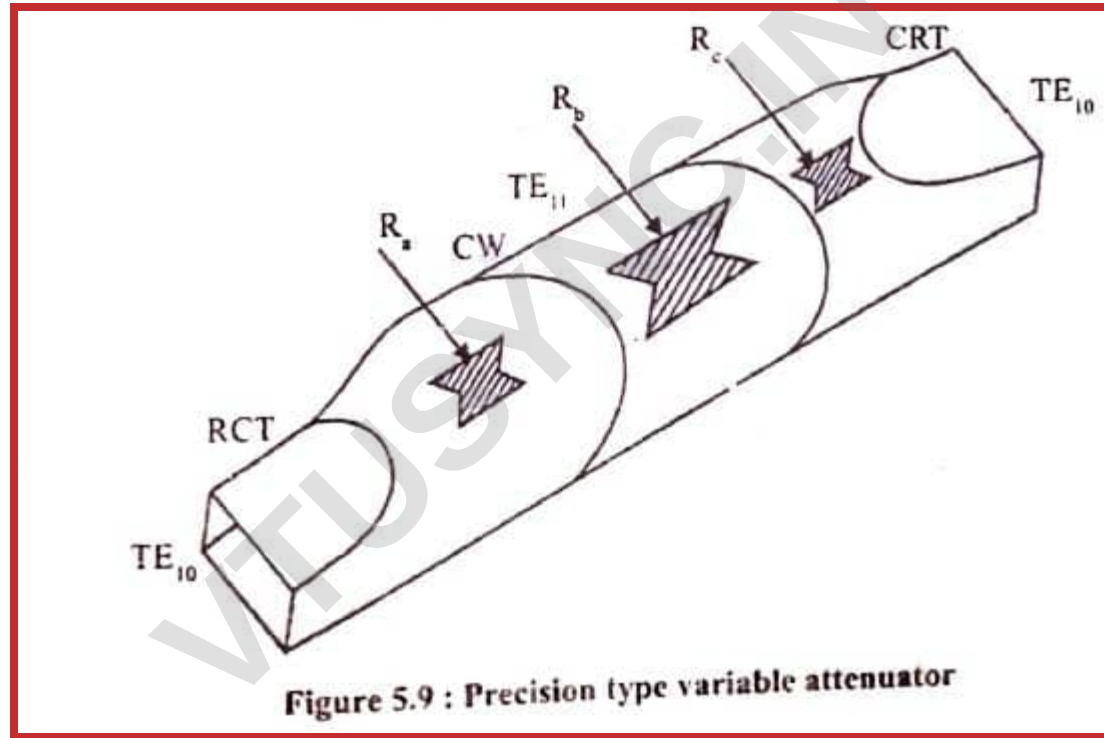
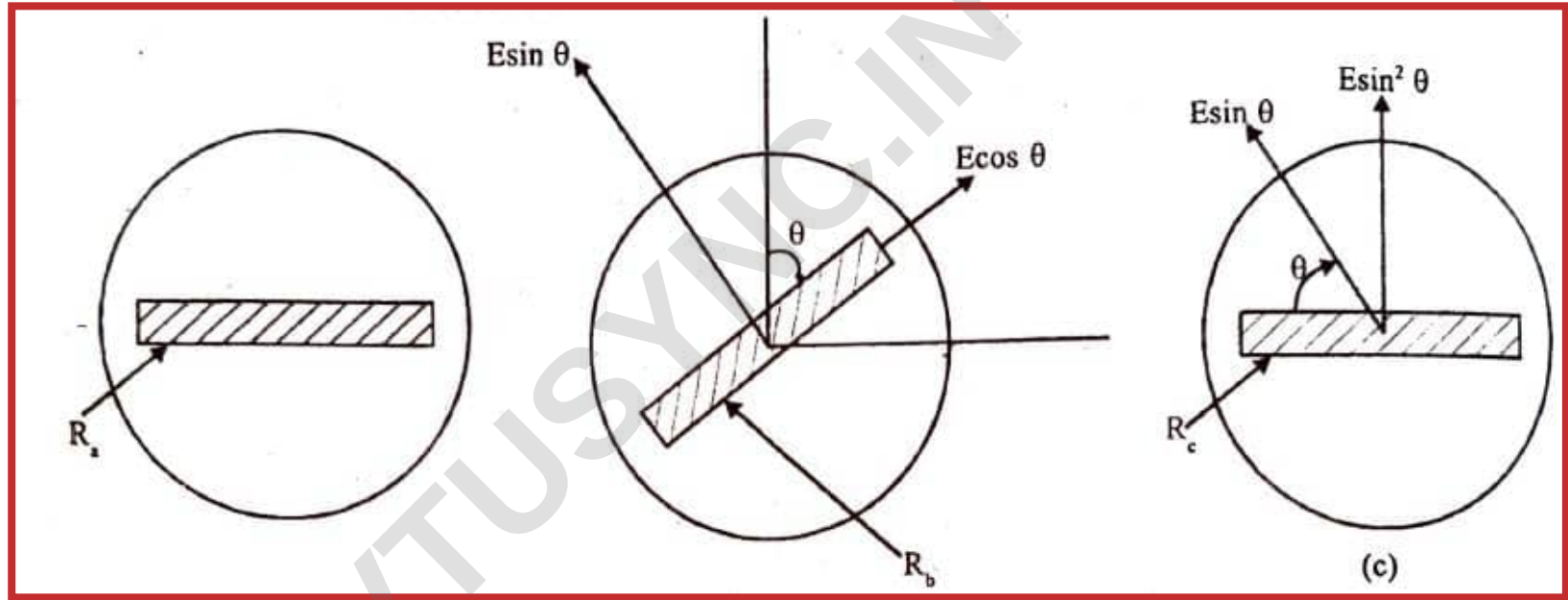


Figure 5.8 shows a flap attenuator which is also a variable attenuator. A semi-circular flap made of lossy dielectric is made to descend into the longitudinal slot cut at the centre of the top wall of rectangular waveguide. When the flap is completely outside the slot, then the attenuation is zero and when it is completely inside, the attenuation is maximum. A maximum of 90 dB attenuation is possible with this attenuator with a VSWR of 1.05. The dielectric slab can be properly shaped according to convenience to get a linear variation of attenuation with depth of insertion.

❖ Precision type variable attenuator



A precision type variable attenuator consists of a rectangular to circular transition (RCT), a piece of circular waveguide (CW) and a circular-to-rectangular transition (CRT) as shown in figure 5.9. Resistive cards R_a , R_b and R_c are placed inside these sections as shown. The centre circular section containing the resistive card R_b can be precisely rotated by 360° with respect to the two fixed resistive cards. The induced current on the resistive card R_b due to the incident signal is dissipated as heat producing attenuation of the transmitted signal. TE_{10} mode in RCT is converted into TE_{11} in circular waveguide. The resistive cards R_a and R_c are kept perpendicular to the electric field of TE_{10} mode so that it does not absorb the energy. But any component parallel to its plane will be readily absorbed. Hence, pure TE_{11} mode is excited in circular waveguide section.



If the resistive card in the centre section is kept at an angle θ relative to the E-field direction of the TE_{11} mode, the component $E \cos\theta$ parallel to the card get absorbed while the component $E \sin \theta$ is transmitted without attenuation. This component finally comes out as $E \sin^2 \theta$ as shown in figure 5.10.

$$\therefore \text{Attenuation } \alpha = \frac{E}{E \sin^2 \theta} = \frac{1}{\sin^2 \theta} = \frac{1}{|S_{21}|} \quad \dots (5.6)$$

Thus the attenuation is a function of " θ " only.

As the attenuator is a reciprocal device

$$|S_{21}| = |S_{12}| \quad \dots (5.7)$$

Since the VSWR is very close to 1 for these attenuations,

$$S_{11} = S_{22} = 0 \quad \dots (5.8)$$

$\therefore$ The S matrix of an ideal precision attenuator is given by

$$[S] = \begin{bmatrix} 0 & \sin^2 \theta \\ \sin^2 \theta & 0 \end{bmatrix} \quad \dots (5.9)$$

PHASE SHIFTERS

A microwave phase shifter is a two port device which produces a variable shift in phase of the incoming microwave signal. A lossless dielectric slab when placed inside the rectangular waveguide as shown in figure 5.11, produces a phase shift.

In fact, a differential phase change is produced due to the change of wave velocity (group velocity) through the dielectric slab compared to that through the empty space inside the waveguide. Two ports are matched reducing the reflections of the wave from the dielectric slab tapered at both ends as shown in figure 5.11.

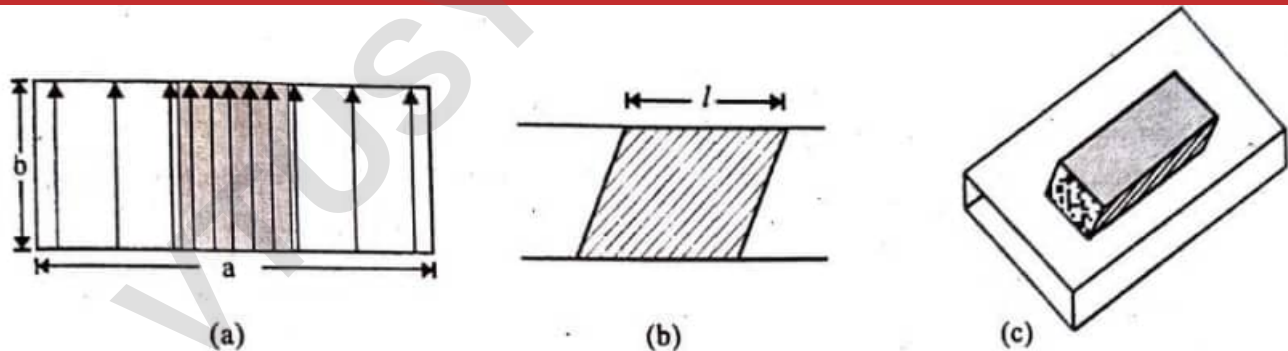


Figure 5.11 : Microwave dielectric phase shifter (a) cross-sectional view (b) side view (c) 3-dimensional view.

“ The phase constant β_d through the dielectric slab is given by

$$\beta_d = \frac{2\pi}{\lambda_{gd}} = \frac{2\pi}{\lambda_o \sqrt{\epsilon_r - \left(\lambda_o/2a\right)^2}} \text{ for TE}_{10} \text{ mode}$$

$$\therefore \beta_d = \frac{2\pi}{\lambda_o} \sqrt{\epsilon_r - \left(\lambda_o/2a\right)^2} \quad \dots (5.10)$$

The phase constant β_o through the empty space is given by

$$\beta_o = \frac{2\pi}{\lambda_g} = \frac{2\pi}{\lambda_o \sqrt{1 - \left(\lambda_o/2a\right)^2}}$$

$$\therefore \beta_o = \frac{2\pi}{\lambda_o} \sqrt{1 - \left(\frac{\lambda_o}{2a}\right)^2} \quad \dots (5.11)$$

Thus the differential phase shift produced by the phase shifter is

$$\Delta\phi = (\beta_d - \beta_0) l$$

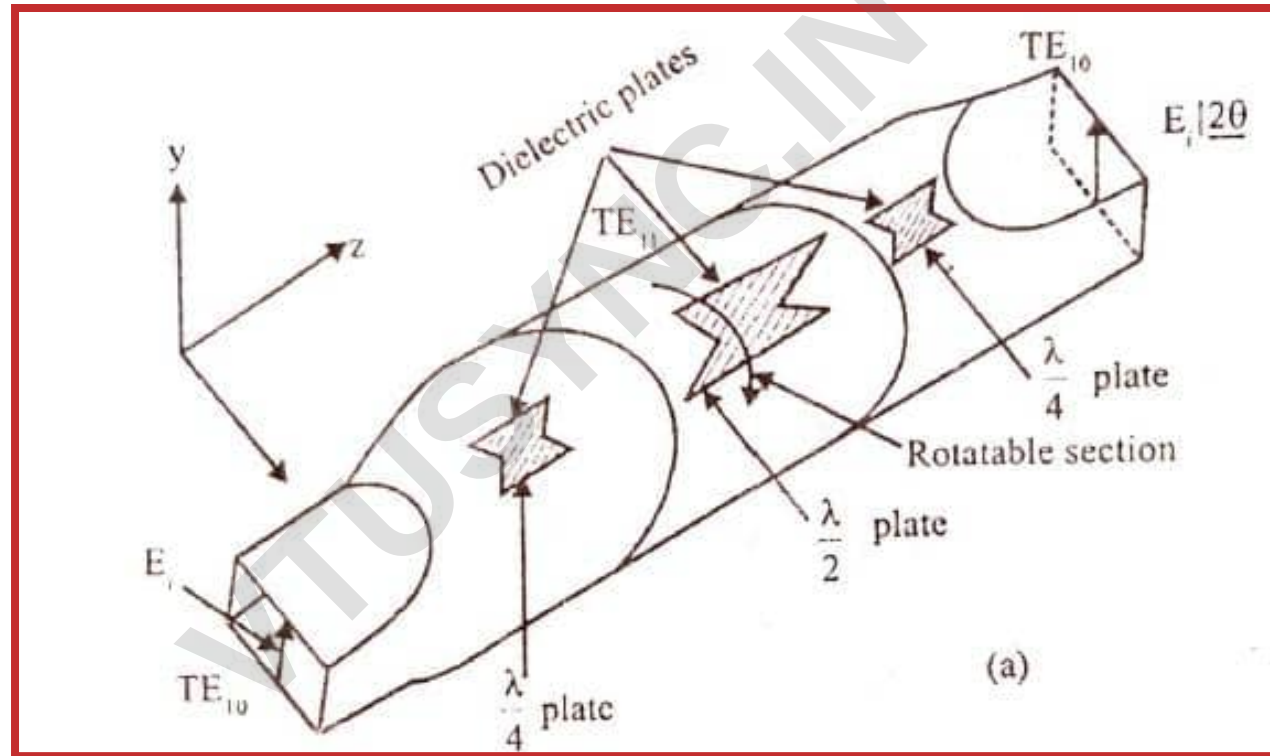
..... (5.12)

By adjusting the length l of the dielectric slab, $\Delta\phi$ can be varied thus getting a variable phase shifter. The S-matrix of an ideal phase shifter is given by

$$[S] = \begin{bmatrix} 0 & e^{-j\Delta\phi} \\ e^{-j\Delta\phi} & 0 \end{bmatrix}$$

..... (5.13)

PRECISION PHASE SHIFTER



The rotary type of precision phase shifter is shown in figure 5.12 which consists of a circular waveguide containing a lossless dielectric plate of length $2l$ called "*half-wave section*", a section of rectangular-to-circular transition containing a lossless dielectric plate of length l , called "*quarter-wave section*", oriented at an angle of 45° to the broader wall of the rectangular waveguide and a circular-to-rectangular transition again containing a lossless dielectric plate of same length l (quarter wave section) oriented at an angle 45° . The incident TE_{10} mode becomes TE_{11} mode in circular waveguide section. The half-wave section produces a phase shift equal to twice that produced by the quarter wave section. The dielectric plates are tapered at both ends to reduce reflections due to discontinuity.

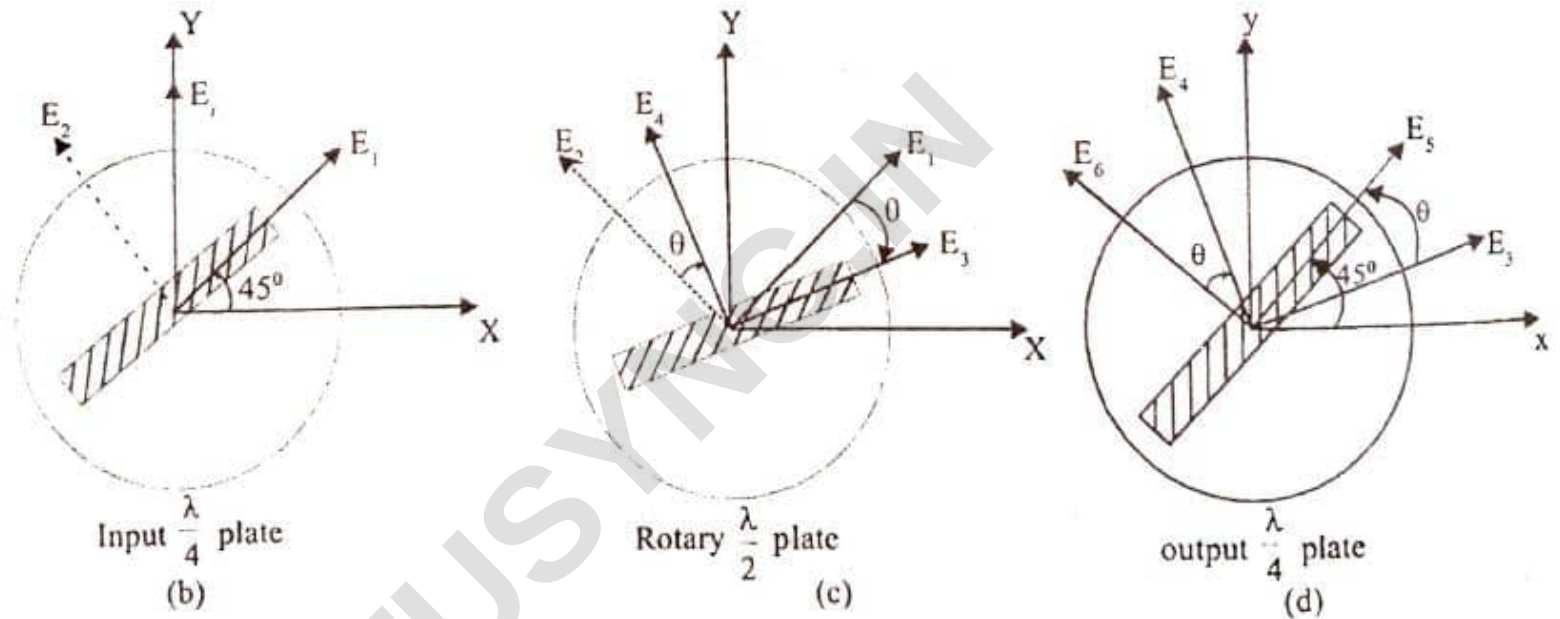


Figure 5.12 : Illustrating precision rotary phase shifter

Let E_i be the maximum electric field strength of the mode which is resolved into two components, E_1 parallel to the plate and E_2 perpendicular to E_1 , as shown in figure 5.12 (b).

After propagation through the plate these components are given by,

$$E_1 = (E_i \cos 45^\circ) e^{-j\beta_1 l} = E_0 e^{-j\beta_1 l} \quad \dots (5.14)$$

$$\text{and} \quad E_2 = (E_i \sin 45^\circ) e^{-j\beta_2 l} = E_0 e^{-j\beta_2 l} \quad \dots (5.15)$$

$$\text{Where} \quad E_0 = \frac{E_i}{\sqrt{2}} \quad \dots (5.16)$$

The length l is adjusted such that these two components E_1 and E_2 have equal amplitude but differing in phase by $(\beta_1 l - \beta_2 l) = 90^\circ$. With this equations (5.14) and (5.15) get modified as

$$E_1 = E_0 e^{-j\beta_1 l} \quad \dots (5.17)$$

$$E_2 = E_0 e^{-j(\beta_1 l - 90^\circ)} = E_0 e^{-j(\beta_1 l - \frac{\pi}{2})}$$

$$\therefore E_2 = E_0 e^{-j\beta_1 l} e^{j\pi/2}$$

$$\therefore E_2 = E_1 e^{j\pi/2} \quad \dots (5.18)$$

As seen from equation (5.17) and (5.18), the quarter wave sections convert a linearly polarized TE_{11} wave into a circularly polarized wave and vice-versa.

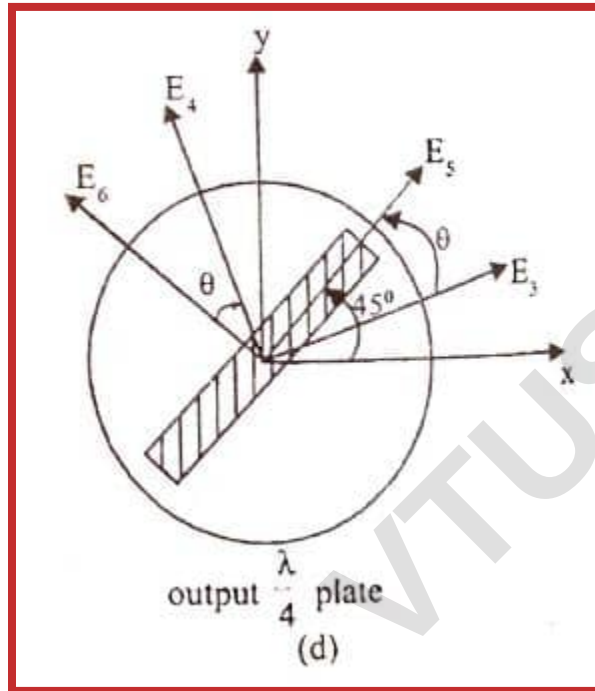
After emerging out of the half-wave section, the electric field components parallel and perpendicular to the half-wave plate are given by

$$\begin{aligned}
 E_3 &= (E_1 \cos \theta - E_2 \sin \theta) e^{-j2\beta_1 l} \quad \text{referring to figure 5.12 (c)} \\
 &= (E_1 \cos \theta - E_1 e^{j\pi/2} \sin \theta) e^{-j2\beta_1 l} \quad \text{by using equation (5.18)} \\
 &= E_1 (\cos \theta - j \sin \theta) e^{-j2\beta_1 l} \quad [\text{since } e^{j\pi/2} = \cos \pi/2 + j \sin \pi/2 = j] \\
 &= E_1 e^{-j\theta} e^{-j2\beta_1 l} \\
 &= E_0 e^{-j\beta_1 l} e^{-j\theta} e^{-j2\beta_1 l} \quad \text{by using equation (5.17)} \\
 E_3 &= E_0 e^{-j\theta} e^{-j3\beta_1 l} \quad \text{..... (5.19)}
 \end{aligned}$$

$\therefore$
and

$$\begin{aligned}
 E_4 &= (E_1 \sin \theta + E_2 \cos \theta) e^{-j2\beta_2 l} \\
 &= (E_1 \sin \theta + E_1 e^{j\pi/2} \cos \theta) e^{-j2\beta_2 l} \quad \text{using equation (5.18)} \\
 &= E_1 (\sin \theta + j \cos \theta) e^{-j2\beta_2 l} \\
 &= jE_1 (\cos \theta - j \sin \theta) e^{-j2(\beta_1 l - \frac{\pi}{2})} \\
 &= jE_1 e^{-j\theta} e^{-j2\pi\beta_1 l} e^{j\pi} \\
 &= E_1 e^{-j\theta} e^{-j2\beta_1 l} e^{j\pi/2} e^{j\pi} \quad [\text{since } j = e^{j\pi/2}] \\
 &= E_0 e^{-j\beta_1 l} e^{-j\theta} e^{-j2\beta_1 l} e^{j3\pi/2} \quad \text{by using equation (5.17)} \\
 E_4 &= E_0 e^{-j\theta} e^{-j3\beta_1 l} e^{j3\pi/2} \quad \text{..... (5.20)}
 \end{aligned}$$

At the output end of this quarterwave plate, the field components parallel and perpendicular to the quarter wave plate, by referring to figure 5.12 (d), can be expressed as



$$E_5 = (E_3 \cos\theta + E_4 \sin\theta) e^{-j\beta_1 l}$$

$$= (E_0 e^{-j\theta} e^{-j3\beta_1 l} \cos\theta + E_0 e^{-j\theta} e^{-j3\beta_1 l} e^{j3\pi/2} \sin\theta) e^{-j\beta_1 l}$$

by using equations (5.19) & (5.20)

$$= E_0 (\cos\theta + e^{j3\pi/2} \sin\theta) e^{-j\theta} e^{-j3\beta_1 l} e^{-j\beta_1 l}$$

$$= E_0 (\cos\theta - j \sin\theta) e^{-j\theta} e^{-4\beta_1 l}$$

$$E_5 = E_0 e^{-j\theta} e^{-j\theta} e^{-j4\beta_1 l}$$

$$E_5 = E_0 e^{-j2\theta} e^{-j4\beta_1 l} \quad \dots (5.21)$$

and

$$\bar{E}_6 = (\bar{E}_4 \cos \theta - E_3 \sin \theta) e^{-j\beta_2 l}$$

$\therefore$

$$E_6 = (E_0 e^{-j\theta} e^{-j3\beta_1 l} e^{j3\pi/2} \cos \theta - E_0 e^{-j\theta} e^{-j3\beta_1 l} \sin \theta) e^{-j\beta_2 l} \quad \text{by using equations (5.19) and (5.20)}$$

$\therefore$

$$\begin{aligned} E_6 &= E_0 (e^{j3\pi/2} \cos \theta - \sin \theta) e^{-j\theta} e^{-j3\beta_1 l} e^{-j(\beta_1 l - \frac{\pi}{2})} \\ &= E_0 (-j \cos \theta - \sin \theta) e^{-j\theta} e^{-j3\beta_1 l} e^{-j\beta_1 l} e^{j\pi/2} \\ &= E_0 (-j) (\cos \theta - j \sin \theta) e^{-j\theta} e^{-j4\beta_1 l} e^{j\pi/2} \\ &= E_0 e^{j3\pi/2} e^{-j\theta} e^{-j\theta} e^{-j4\beta_1 l} e^{j\pi/2} \\ &= E_0 e^{-j2\theta} e^{-j4\beta_1 l} e^{j2\pi} \end{aligned}$$

since $e^{j2\pi} = 1$, we get

$$E_6 = E_0 e^{-j2\theta} e^{-j4\beta_1 l} \quad \text{..... (5.22)}$$

Comparison of equation (5.21) and (5.22) yields that the components E_5 and E_6 are identical in both magnitude and phase and the resultant electric field strength at the output is given by

$$\begin{aligned} E_{\text{out}} &= \sqrt{(E_5)^2 + (E_6)^2} \\ &= \sqrt{2} E_0 e^{-j2\theta} e^{-j4\beta_1 l} \end{aligned} \quad \dots (5.23)$$

From equation (5.16) we have $\sqrt{2} E_0 = E_i$

$$\therefore E_{\text{out}} = E_i e^{-j(2\theta + 4\beta_1 l)} \quad \dots (5.24)$$

This equation (5.24) suggests that the polarization of E_{out} and E_i are same except for a phase change of

$$\Delta\phi = 2\theta + 4\beta_1 l \text{ radians} \quad \dots (5.25)$$

At a given frequency, " $4\beta_1 l$ " remains constant and change in phase is achieved by varying angle θ of the half-wave plate with respect to quarter wave plate.

WAVEGUIDE TEES



A waveguide tee junction is formed when three waveguides are interconnected in the form of English Alphabet T and thus waveguide tee is a 3-port junction. The waveguide tees are used to connect a branch or section of waveguide in series or parallel with the main waveguide transmission line for either splitting or combining power in a waveguide system.

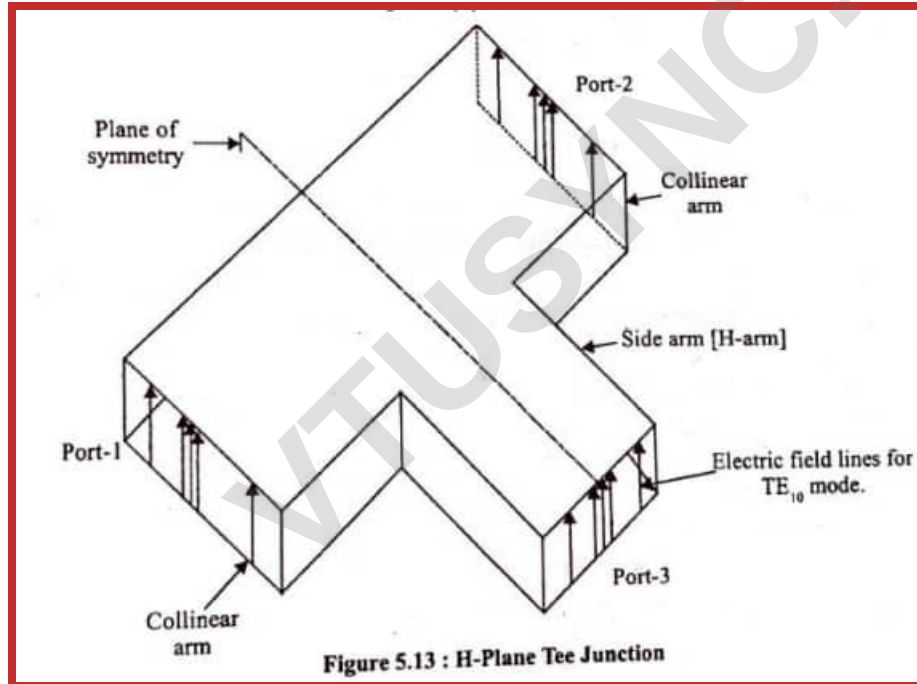
There are basically two types of tees namely

- (i) H-plane Tee junction → Here the side arm is connected so that its axis is parallel to H-field
- (ii) E-plane Tee junction → Here the side arm is connected so that its axis is parallel to E-field

A combination of these two junction tee junctions is also possible. It is called "**E-H plane tee junction**" or "**Hybrid-Tee Junction**". A special case of this junction is called "**MAGIC-TEE**"

(i) H-plane Tee Junction

A H-plane tee junction is formed when a rectangular slot is cut along the width of a long waveguide and a side-arm attached into the slot as shown in figure 5.13. Ports (1) and (2) are called "*collinear arms or ports*" and port-(3) as the "*H-arm*" or "*sum-arm*".



Characteristics of H-plane Tee Junction

1. Because of plane of symmetry, microwave power fed into port-(3) in the dominant mode divides equally between the ports-(1) and (2). Therefore, the scattering coefficients S_{31} and S_{32} are equal both in magnitude and phase. (5.27)

$$\therefore S_{31} = S_{32}$$

2. Let us assume that port-(3) is perfectly matched to the junction i.e., when microwave power is fed into port-(3), nothing comes out of port-(3) reflected from the junction. (5.28)

$$\therefore S_{33} = 0$$

3. From symmetric property of $[S]$ matrix [refer to equation (4.28)], we have

$$S_{ij} = S_{ji} \quad \text{..... (5.29)}$$

With the above 3-characteristics, the $[S]$ matrix of equation (5.26) becomes

$$[S]_{11} = \begin{bmatrix} S_{11} & S_{12} & S_{13} \\ S_{12} & S_{22} & S_{13} \\ S_{13} & S_{13} & S_0 \end{bmatrix} \quad \text{..... (5.30)}$$

$$[\text{Since } S_{21} = S_{12} \text{ and } S_{23} = S_{32} = S_{31} = S_{13}]$$

4. From unitary property [refer equation (4.73)], we have

$$[S][S]^* = [U]$$

Where $[S]^*$ = complex conjugate of $[S]$

$[U]$ = unit matrix of same order as $[S]$

Substituting for S and U we get

$$\begin{bmatrix} S_{11} & S_{12} & S_{13} \\ S_{12} & S_{22} & S_{13} \\ S_{13} & S_{13} & 0 \end{bmatrix} \begin{bmatrix} S_{11}^* & S_{12}^* & S_{13}^* \\ S_{12}^* & S_{22}^* & S_{13}^* \\ S_{13}^* & S_{13}^* & 0 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \quad \text{..... (5.31)}$$

Multiplying the matrices and equating the corresponding equations to the elements of unit matrix, we can get nine equations. Since there are only four unknown elements, only four equations would suffice.

Considering the product of first row elements of 1st matrix with first column elements of the 2nd matrix, we get

$$\begin{aligned} S_{11} S_{11}^* + S_{12} S_{12}^* + S_{13} S_{13}^* &= 1 \\ \therefore |S_{11}|^2 + |S_{12}|^2 + |S_{13}|^2 &= 1 \end{aligned} \quad \text{..... (5.32)}$$

Considering 2nd row of 1st matrix and 2nd column of 2nd matrix, we get

$$S_{12} S_{12}^* + S_{22} S_{22}^* + S_{13} S_{13}^* = 1$$

$$\therefore |S_{12}|^2 + |S_{22}|^2 + |S_{13}|^2 = 1 \quad \text{..... (5.33)}$$

Similarly, the 3rd row of 1st matrix and 3rd column of 2nd matrix yields

$$S_{13} S_{13}^* + S_{13} S_{13}^* + 0 = 1$$

$$\therefore |S_{13}|^2 + |S_{13}|^2 = 1$$

$$\text{or } 2|S_{13}|^2 = 1 \quad \text{..... (5.34)}$$

To get the fourth equation, let us consider the product of 3rd row of 1st matrix and first column of 2nd matrix and equating to zero, we get

$$S_{13} S_{11}^* + S_{13} S_{12}^* = 0 \quad \text{..... (5.35)}$$

From equation (5.34),

$$S_{13} = \frac{1}{\sqrt{2}} \quad \text{..... (5.36)}$$

Comparing equations (5.32) and (5.33), we get

$$S_{11} = S_{22} \quad \text{..... (5.37)}$$

From equation (5.35),

$$S_{13} (S_{11}^* + S_{12}^*) = 0$$

Since $S_{13} \neq 0$, we must have $S_{11}^* + S_{12}^* = 0$

$\therefore$

$$S_{12}^* = -S_{11}^*$$

Removing complex conjugate on both sides,

$$S_{12} = -S_{11}$$

..... (5.38)

Using equations (5.38) and (5.36) in equation (5.32), we get

$$|S_{11}|^2 + |S_{11}|^2 + \left(\frac{1}{\sqrt{2}}\right)^2 = 1$$

$\therefore$

$$2 |S_{11}|^2 = \frac{1}{2}$$

$\therefore$

$$S_{11} = \frac{1}{2}$$

..... (5.39)

$\therefore$

$$S_{12} = -S_{11} = -\frac{1}{2}$$

..... (5.40)

$\therefore$

$$S_{22} = S_{11} = \frac{1}{2}$$

..... (5.41)

Using these values of S parameters [equations (5.36), (5.39) to (5.41)] in equation (5.30), we get

Using these values of S parameters [equations (5.36), (5.39) to (5.41)] in equation (5.30), we get

$$[S]_H = \begin{bmatrix} \frac{1}{2} & -\frac{1}{2} & \frac{1}{\sqrt{2}} \\ -\frac{1}{2} & \frac{1}{2} & \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} & 0 \end{bmatrix} \quad \dots (5.42)$$

This matrix is the [S] matrix of H-plane Tee junction with port(3) perfectly matched to the junction.

we have from equation (4.19),

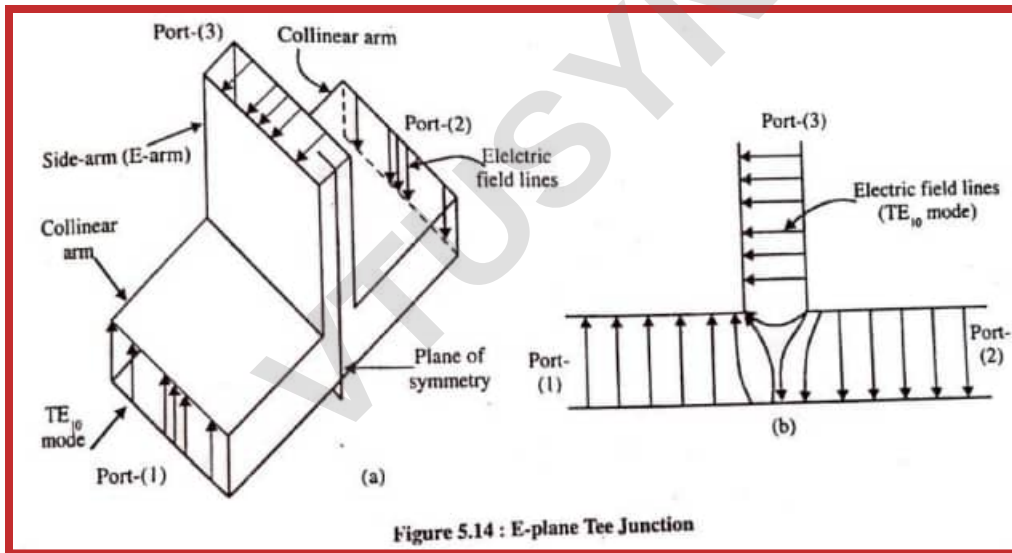
$$[b] = [S]_H [a]$$

$$\begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix} = \begin{bmatrix} \frac{1}{2} & -\frac{1}{2} & \frac{1}{\sqrt{2}} \\ -\frac{1}{2} & \frac{1}{2} & \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} & 0 \end{bmatrix} \begin{bmatrix} a_1 \\ a_2 \\ a_3 \end{bmatrix}$$

(ii) E-Plane Tee Junction:

A E-plane tee junction is formed when a rectangular slot is cut along the breadth of a long waveguide and a side-arm is inserted from top as shown in figure 5.14 (a).

When microwave signal in TE_{10} mode is allowed to propagate through the E-arm, the electric field lines turn at the corner reorienting themselves in such a way that a phase shift of 180° is produced between the electric field lines coming out of ports-(1) and (2), as shown in figure 5.14 (b).





Characteristics of E-plane Tee-Junction:

1. Because of plane of symmetry, microwave power fed into port-(3) in the dominant TE_{10} mode, divides equally between the ports (1) and (2), but with a phase shift of 180° between them. Therefore, the scattering coefficients S_{31} and S_{32} are equal in magnitude but opposite in phase,

$$\dots (5.49)$$

$$\text{i.e.,} \quad S_{32} = -S_{31}$$

2. Let us again assume that port-(3) is perfectly matched to the junction.

$$\dots (5.50)$$

$$\therefore \quad S_{33} = 0$$

3. From symmetric property of $[S]$ matrix, we have

$$\dots (5.51)$$

$$S_{ij} = S_{ji}$$

With the above three characteristics, the $[S]$ matrix of equation (5.26) becomes

$$[S] = \begin{bmatrix} S_{11} & S_{12} & S_{13} \\ S_{12} & S_{22} & -S_{13} \\ S_{13} & -S_{13} & 0 \end{bmatrix}$$

$$\dots (5.52)$$

$$[\text{Since } S_{21} = S_{12}, S_{23} = S_{32} = -S_{31} = -S_{13} \text{ and } S_{31} = S_{13}]$$

4. From the unitary property of equation (4.73), we have

$$[S][S]^* = [U]$$

$$\begin{bmatrix} S_{11} & S_{12} & S_{13} \\ S_{12} & S_{22} & -S_{13} \\ S_{13} & -S_{13} & 0 \end{bmatrix} \begin{bmatrix} S_{11}^* & S_{12}^* & S_{13}^* \\ S_{12}^* & S_{22}^* & -S_{13}^* \\ S_{13}^* & -S_{13}^* & 0 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \quad \dots (5.53)$$

Here also we can obtain nine equations, but only four would suffice.

Considering the product of 1st row & 1st column, we get

$$|S_{11}|^2 + |S_{12}|^2 + |S_{13}|^2 = 1 \quad \dots (5.54)$$

The product of 2nd row & 2nd column yields

$$|S_{12}|^2 + |S_{22}|^2 + |S_{13}|^2 = 1 \quad \dots (5.55)$$

The multiplication of 3rd row with 3rd column elements results in

$$|S_{13}|^2 + |S_{13}|^2 + 0 = 1 \quad \dots (5.56)$$

And finally considering 3rd row and 1st column elements, we obtain

$$S_{13} S_{11}^* - S_{13} S_{12}^* + 0 = 0 \quad \dots (5.57)$$

From equation (5.56), we get

$$2|S_{13}|^2 = 1$$

$$\therefore S_{13} = \frac{1}{\sqrt{2}} \quad \text{..... (5.58)}$$

Comparison of equations (5.54) and (5.55) yields

$$S_{22} = S_{11} \quad \text{..... (5.59)}$$

From equation (5.57), we have

$$S_{13}(S_{11}^* - S_{12}^*) = 0$$

Since $S_{13} \neq 0$, we must have $S_{11}^* - S_{12}^* = 0$

$$\therefore S_{11}^* = S_{12}^*$$

Taking out complex conjugates, $S_{12} = S_{11}$ (5.60)

Using equations (5.58) and (5.60) in (5.54), we get

$$|S_{11}|^2 + |S_{11}|^2 + \frac{1}{2} = 1$$

$$\therefore 2|S_{11}|^2 = \frac{1}{2}$$

$$\therefore S_{11} = \frac{1}{2} = S_{12} = S_{22} \quad \text{..... (5.61)}$$

Using these values of S-parameters of equations (5.58) and (5.61) in the matrix of equation (5.52), we get the [S] matrix of E-plane Tee junction as

$$[S]_E = \begin{bmatrix} \frac{1}{2} & \frac{1}{2} & \frac{1}{\sqrt{2}} \\ \frac{1}{2} & \frac{1}{2} & -\frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} & 0 \end{bmatrix} \quad \dots (5.62)$$

(iii) E-H PLANE TEE JUNCTION (HYBRID-TEE)

A E-H plane-Tee junction or Hybrid-tee junction is formed when E and H plane-tee junctions are combined as shown in figure 5.15. Rectangular slots are cut both along the width and breadth of a long waveguide and side-arms are attached into it. In figure 5.15, ports (1) and (2) form the collinear ports, port (3) the H-arm and port (4) the E-arm.

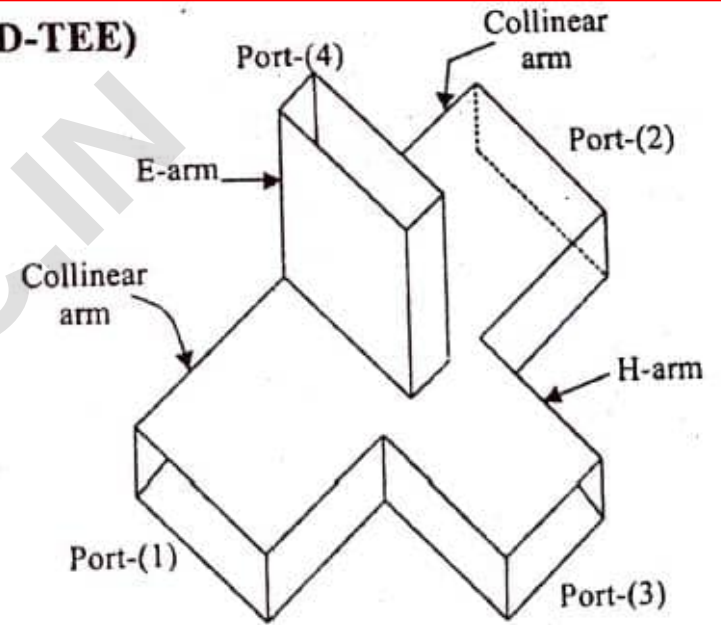


Figure 5.15 : E-H plane-tee or Hybrid-Tee Junction



Seventh Semester B.E. Degree Examination, Dec.2018/Jan.2019

Microwaves and Antennas

Time: 3 hrs.

Max. Marks: 80

Note: Answer any FIVE full questions, choosing
ONE full question from each module.

Module-1

- 1 a. Derive the general transmission line equation to find voltage and current on the line in terms of position 'z' and time 't'. (07 Marks)
- b. Describe the different mode curve in the case of reflex klystron. (05 Marks)
- c. A transmission line has a characteristic impedance of $50 + j0.01\Omega$ and terminated in a load impedance of $73 - j42.5\Omega$ calculate : i) reflection coefficient ii) SWR. (04 Marks)

OR

- 2 a. Define reflection coefficient. Derive the equation for reflection coefficient at the load end at a distance 'd' from the load. (06 Marks)
- b. Describe the mechanism of oscillation of reflex klystron. (06 Marks)
- c. A transmission line has the following parameters : $R = 2\Omega/m$, $G = 0.5\text{mmho/m}$, $f = 1\text{GHz}$, $L = 8\text{nH/m}$, $C = 0.23\text{pF/m}$. Calculate : i) characteristic impedance ii) propagation constant. (04 Marks)

Module-2

- 3 a. State and explain the properties of S - matrix. (07 Marks)
- b. With a neat diagram, explain the working of precession type variable attenuator. (06 Marks)
- c. A 20mW signal is fed into one of the collinear port 1 of a lossless H-plane T junction. Calculate the power delivered through each port when other ports are terminated in matched load. (03 Marks)

OR

- 4 a. What is magic Tee? Derive its scattering matrix. (06 Marks)
- b. Discuss different types of coaxial connectors. (04 Marks)
- c. 2 transmission lines of characteristic impedance Z_1 and Z_2 are joined at plane PP'. Express S-parameters in terms of impedance when each line is matched terminated. (06 Marks)

Seventh Semester B.E. Degree Examination, Dec.2019/Jan.2020

Microwaves and Antennas

Time: 3 hrs.

Max. Marks: 80

Note: Answer any FIVE full questions, choosing ONE full question from each module.

Module-1

- 1
 - a. List four applications of Reflex Klystron. (04 Marks)
 - b. Derive transmission line equations in voltage and current forms. (06 Marks)
 - c. A transmission line is terminated in a resistive load of 1000Ω and has $L = 9\mu\text{H/m}$ and $C = 100\text{pF/m}$. Calculate reflection coefficient and standing wave ratio. (06 Marks)

OR

- 2
 - a. Define reflection coefficient. Derive an expression for reflection coefficient at load in terms of characteristic impedance and load impedance. (08 Marks)
 - b. Explain microwave system with the aid of a diagram. (08 Marks)

Module-2

- 3
 - a. For a two port network with mismatched load derive an expression for input reflection coefficient. (06 Marks)
 - b. Draw the diagram of Magic-Tee. Derive S-matrix of Magic Tee. (10 Marks)

Characteristics of E-H plane Tee Junction:-

1. E-H plane tee is a 4-port junction and therefore the scattering matrix is of the order 4×4 given by

$$[S]_{EH} = \begin{bmatrix} S_{11} & S_{12} & S_{13} & S_{14} \\ S_{21} & S_{22} & S_{23} & S_{24} \\ S_{31} & S_{32} & S_{33} & S_{34} \\ S_{41} & S_{42} & S_{43} & S_{44} \end{bmatrix} \quad \dots (5.68)$$

2. Because of H-plane tee action, $S_{32} = S_{31}$ (5.69)

3. Because of E-plane tee action, $S_{42} = -S_{41}$ (5.70)

4. Because of geometry of the junction, the microwave signal fed into port-(3) cannot induce the dominant mode in port-(4) and vice-versa. Therefore, the ports(3) and (4) behave as "**ISOLATED PORTS**". The scattering coefficients S_{34} and S_{43} will be zero.

$$\therefore S_{34} = S_{43} = 0 \quad \dots (5.71)$$

5. Let us assume that E-arm and H-arm are perfectly matched to the junction

$$\therefore S_{33} = 0 \text{ and } S_{44} = 0 \quad \dots (5.72)$$

6. From symmetric property, we have $S_{ij} = S_{ji}$

With the above characteristics, the scattering matrix of equation (5.68) becomes

$$[S]_{EH} = \begin{bmatrix} S_{11} & S_{12} & S_{13} & S_{14} \\ S_{12} & S_{22} & S_{13} & -S_{14} \\ S_{13} & S_{13} & 0 & 0 \\ S_{14} & -S_{14} & 0 & 0 \end{bmatrix} \quad \dots (5.73)$$

[Since $S_{21} = S_{12}$, $S_{23} = S_{32} = S_{31} = S_{13}$, $S_{24} = S_{42} = -S_{41} = -S_{14}$ and $S_{41} = S_{14}$]

7. From the unitary property $[S][S]^* = [U]$

$$\begin{bmatrix} S_{11} & S_{12} & S_{13} & S_{14} \\ S_{12} & S_{22} & S_{13} & -S_{14} \\ S_{13} & S_{13} & 0 & 0 \\ S_{14} & -S_{14} & 0 & 0 \end{bmatrix} \begin{bmatrix} S_{11}^* & S_{12}^* & S_{13}^* & S_{14}^* \\ S_{12}^* & S_{22}^* & S_{13}^* & -S_{14}^* \\ S_{13}^* & S_{13}^* & 0 & 0 \\ S_{14}^* & -S_{14}^* & 0 & 0 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \quad \dots (5.74)$$

Considering 1st row and 1st column, we get,

$$|S_{11}|^2 + |S_{12}|^2 + |S_{13}|^2 + |S_{14}|^2 = 1 \quad \dots (5.75)$$

Considering 1st row and 1st column, we get,

$$|S_{11}|^2 + |S_{12}|^2 + |S_{13}|^2 + |S_{14}|^2 = 1 \quad \dots (5.75)$$

Considering 2nd row and 2nd column, we get

$$|S_{12}|^2 + |S_{22}|^2 + |S_{13}|^2 + |S_{14}|^2 = 1 \quad \dots (5.76)$$

Considering 3rd row and 3rd column, we get

$$|S_{13}|^2 + |S_{13}|^2 + 0 + 0 = 1 \quad \dots (5.77)$$

Considering 4th row and 4th column, we get

$$|S_{14}|^2 + |S_{14}|^2 + 0 + 0 = 1 \quad \dots (5.78)$$

From equation (5.77), $2|S_{13}|^2 = 1 \quad \therefore S_{13} = \frac{1}{\sqrt{2}}$ (5.79)

From equation (5.78), $2|S_{14}|^2 = 1 \quad \therefore S_{14} = \frac{1}{\sqrt{2}}$ (5.80)

Comparison of equations (5.75) and (5.76) yields

$$S_{22} = S_{11} \quad \dots (5.81)$$

Using the values obtained from equations (5.79) and (5.80) in (5.75), we get

$$|S_{11}|^2 + |S_{12}|^2 + \frac{1}{2} + \frac{1}{2} = 1$$

$$\therefore |S_{11}|^2 + |S_{12}|^2 = 0 \quad \text{..... (5.82)}$$

This equation (5.82) has only one solution that both of them must be zero.

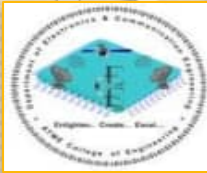
$$\therefore S_{11} = S_{12} = 0 \quad \text{..... (5.83)}$$

From equation (5.81),

$$S_{22} = S_{11} = 0 \quad \text{..... (5.84)}$$

Using the values of S-parameters obtained in equations (5.79), (5.80), (5.83) and (5.84) in the matrix of equation (5.73), we get the scattering matrix of magic-tee as

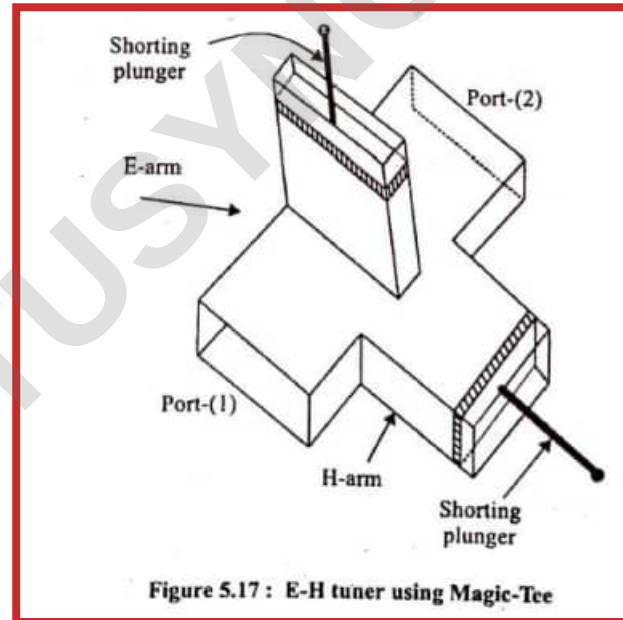
$$[S]_{MT} = \begin{bmatrix} 0 & 0 & \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ 0 & 0 & \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} & 0 & 0 \\ \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} & 0 & 0 \end{bmatrix} \quad \text{..... (5.85)}$$



Applications of Magic-Tee:

I. E-H tuner for impedance matching

Figure 5.17 shows a E-H tuner with shorting plates inserted into the E and H-arms of a magic-tee. These shorting plates act as E and H-plane stubs. The position of these shorting plungers can be adjusted so that a wide range of load impedances may be matched to reduce the VSWR of a waveguide system connected through the collinear arms.



2. Magic-tee used as balanced mixer

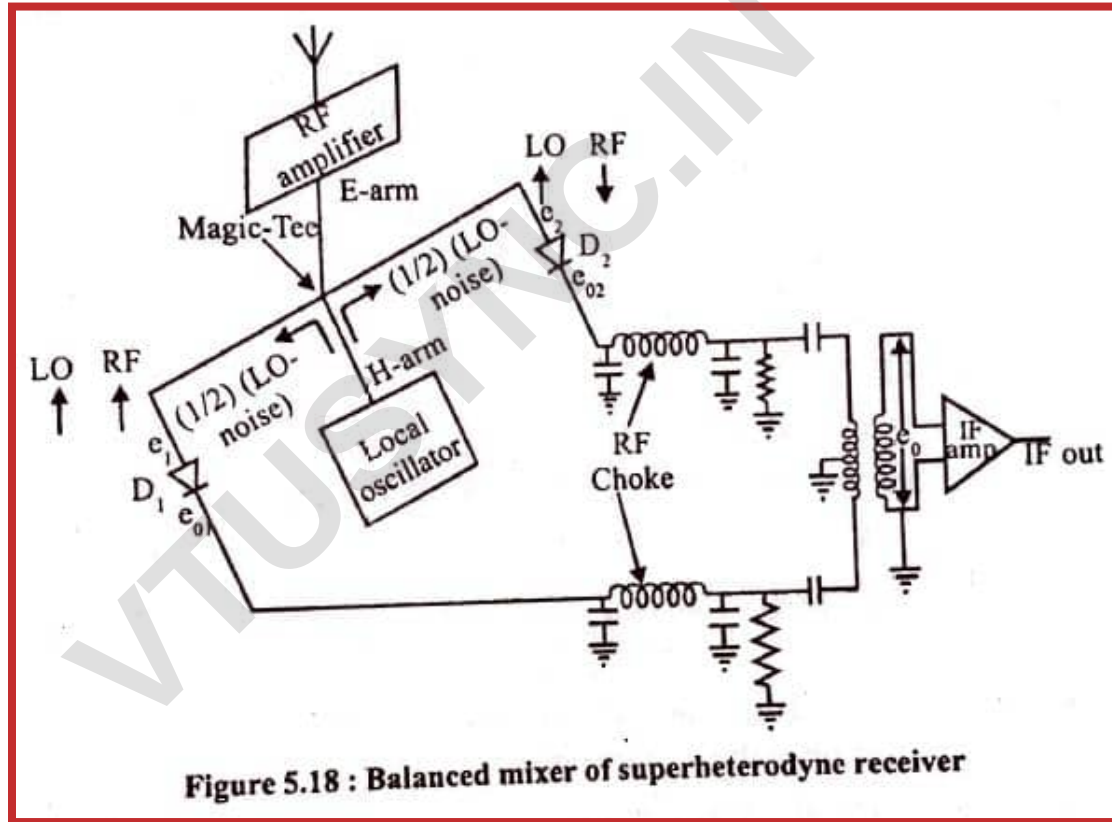


Figure 5.18 shows a balanced mixer in a superheterodyne receiver configuration. The local oscillator is connected to the H-arm and the RF signal from the antenna is amplified and applied to the E-arm of a magic-tee. The crystal diodes D_1 and D_2 , used as mixer diodes, are connected to the collinear ports (1) and (2) respectively. The local oscillator signal "LO" and RF signal shown as "RF", will be in phase at port-(1) (designated in figure 5.18 as LO $\uparrow$ and RF $\uparrow$) and in phase opposition at port-(2) (shown in figure 5.18 as LO $\uparrow$ and RF $\downarrow$). The non-linear characteristics of the diodes will be used for the mixing action to produce the intermediate frequency $f_{IF} = f_{LO} - f_{RF}$. The noise produced by the local oscillator also gets divided between the collinear ports (1) and (2) and will be in phase at the position of the two diodes. The diode outputs are then subtracted to cancel out this noise and only the pure IF signal is amplified by the IF amplifier. The local oscillator signal and the RF signal are not coupled as E and H-arms of magic-tee are isolated ports.

Example 5.1 : Consider a lossless H-plane Tee junction with 50 mW of power being fed into port (1) and the other two ports (2) and (3) are terminated in matched termination. Calculate the power fed into each of the ports by the junction.

$$[S]_H = \begin{bmatrix} \frac{1}{2} & -\frac{1}{2} & \frac{1}{\sqrt{2}} \\ -\frac{1}{2} & \frac{1}{2} & \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} & 0 \end{bmatrix}$$

Seventh Semester B.E. Degree Examination, Dec.2018/Jan.2019
Microwaves and Antennas

Time: 3 hrs.

Max. Marks: 80

**Note: Answer any FIVE full questions, choosing
ONE full question from each module.**

Module-1

- 1 a. Derive the general transmission line equation to find voltage and current on the line in terms of position 'z' and time 't'. (07 Marks)
b. Describe the different mode curve in the case of reflex klystron. (05 Marks)
c. A transmission line has a characteristic impedance of $50 + j0.01\Omega$ and terminated in a load impedance of $73 - j42.5\Omega$ calculate : i) reflection coefficient ii) SWR. (04 Marks)

OR

- 2 a. Define reflection coefficient. Derive the equation for reflection coefficient at the load end at a distance 'd' from the load. (06 Marks)
b. Describe the mechanism of oscillation of reflex klystron. (06 Marks)
c. A transmission line has the following parameters : $R = 2\Omega/m$, $G = 0.5\text{mmho/m}$, $f = 1\text{GHz}$, $L = 8\text{nH/m}$, $C = 0.23\text{pF/m}$. Calculate : i) characteristic impedance ii) propagation constant. (04 Marks)

Module-2

- 3 a. State and explain the properties of S - matrix. (07 Marks)
b. With a neat diagram, explain the working of precession type variable attenuator. (06 Marks)
c. A 20mW signal is fed into one of the collinear port 1 of a lossless H-plane T junction. Calculate the power delivered through each port when other ports are terminated in matched load. (03 Marks)

Solution : From S matrix of H-plane Tee junction of equation (5.42), we have $S_{11} = \frac{1}{2}$,

$$S_{12} = -\frac{1}{2} \text{ and } S_{13} = \frac{1}{\sqrt{2}}.$$

Power fed into port (1) = $P_{in} = 50 \text{ mW}$

$\therefore$ Power reflected back by the junction due to mismatch which comes out of port (1)

$$\begin{aligned} P_1 &= P_{in} |S_{11}|^2 \\ &= (50) \left(\frac{1}{2}\right)^2 = 12.5 \text{ mW} \end{aligned}$$

$$\therefore P_1 = 12.5 \text{ mW}$$

$$\begin{aligned} \text{Power coming out of port (2)} = P_2 &= P_{in} |S_{12}|^2 \\ &= (50) \left|-\frac{1}{2}\right|^2 \end{aligned}$$

$$\therefore P_2 = 12.5 \text{ mW}$$

$$\begin{aligned} \text{And power coming out of port (3)} = P_3 &= P_{in} |S_{13}|^2 \\ &= (50) \left(\frac{1}{\sqrt{2}}\right)^2 \end{aligned}$$

$$\therefore P_3 = 25 \text{ mW}$$

Thus $P_{in} = P_1 + P_2 + P_3$ as the junction is lossless.

Example 5.2 : Consider a lossless H-plane Tee junction with 40 mW power being fed into a perfectly matched port (3) of the junction at $50\ \Omega$. The two collinear ports (1) and (2) are terminated in mismatched loads of $45\ \Omega$ and $55\ \Omega$ respectively. Calculate the actual powers delivered to these loads and the corresponding reflected powers.

Solution : The S-matrix of H-plane Tee junction is given by equation (5.42) as [with port (3) matched].

$$[S]_H = \begin{bmatrix} \frac{1}{2} & -\frac{1}{2} & \frac{1}{\sqrt{2}} \\ -\frac{1}{2} & \frac{1}{2} & \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} & 0 \end{bmatrix}$$

Seventh Semester B.E. Degree Examination, June/July 2019
Microwave and Antennas

Time: 3 hrs.

Max. Marks: 80

Note: Answer any FIVE full questions, choosing ONE full question from each module.

Module-1

1.
 - a. Discuss mechanism of oscillation in Reflex Klystron with schematic. (06 Marks)
 - b. A Reflex Klystron is to be operated at 10GHz with dc beam voltage 300V, repeller space 0.1cm for $1\frac{3}{4}$ mode. Calculate P_{RFmax} and corresponding repeller voltage for a beam current of 20mA. (05 Marks)
 - c. A transmission line has the following parameters:
 $R = 2\Omega/m$, $G = 0.5mho/m$, $f = 1GHz$, $L = 8nH/m$ and $C = 0.23PF$.
 Calculate its characteristics impedance and propagation constant. (05 Marks)

OR

2.
 - a. A line of 400Ω is connected to a load of $200 + j300\Omega$ which is excited by a matched generator at 800MHz. Find the location and length of a single stub nearest to the load to produce an impedance match. (08 Marks)
 - b. A certain transmission line has a characteristic impedance of $75 + j0.01\Omega$ and is terminated in a load impedance of $75 + j50\Omega$. Compute: i) Reflection coefficient ii) The transmission coefficient. (04 Marks)
 - c. What are the high frequency limitations of conventional vacuum tube / transistors? (04 Marks)

Module-2

3.
 - a. Show that impedance and admittance matrices are symmetrical for a reciprocal junction. (06 Marks)
 - b. In a H-plane T-junction, compute power delivered to the loads 40ohm and 60ohm connected to arms 1 and 2 when 10mw power is delivered to matched port 3. Assume characteristics impedance of line = 50ohm. (04 Marks)

$$[b] = [S]_H [a].$$

As there are no inputs at ports (1) and (2), we have $a_1 = a_2 = 0$.

$$\therefore \begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix} = \begin{bmatrix} \frac{1}{2} & -\frac{1}{2} & \frac{1}{\sqrt{2}} \\ -\frac{1}{2} & \frac{1}{2} & \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} & 0 \end{bmatrix} \begin{bmatrix} 0 \\ 0 \\ a_3 \end{bmatrix}$$

$$\therefore b_1 = \frac{1}{\sqrt{2}} a_3 \quad \text{and} \quad b_2 = \frac{1}{\sqrt{2}} a_3$$

Thus the input power $P_3 = 40 \text{ mW}$ divides equally between ports (1) and (2).

$\therefore$ Power going into port (1) and port (2) = 20 mW

Power reflected back by port (1) due to mismatched load of $45 \Omega = \frac{1}{2} |K_1 a_3|^2$

where $|K_1|$ = reflection coefficient at port (1)

$$= \left| \frac{45 - 50}{45 + 50} \right| = \frac{1}{19}$$

$$\therefore \text{Reflected power at port (1)} = \frac{1}{2} |a_3|^2 |K_1|^2$$

$$= (20 \text{ mW}) \left(\frac{1}{19} \right)^2$$

$$= 0.0554 \text{ mW}$$

$$\therefore \text{Actual power delivered to } 45 \Omega \text{ load at port (1)} = 20 \text{ mW} - 0.0554 \text{ mW} = 19.9446 \text{ mW}$$

Similarly, power reflected back by port (2) due to mismatched load of $55 \Omega = \frac{1}{2} |K_2 a_3|^2$

where

$|K_2|$ = reflection coefficient at port (2)

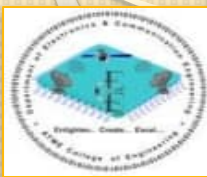
$$= \left| \frac{55 - 50}{55 + 50} \right|$$

$$|K_2| = \frac{1}{21}$$

$$\begin{aligned} \therefore \text{Reflected power at port (2)} &= \frac{1}{2} |a_3|^2 |K_2|^2 \\ &= (20 \text{ mW}) \left(\frac{1}{21} \right)^2 \\ &= \mathbf{0.0454 \text{ mW}} \end{aligned}$$

Actual power delivered to 55Ω load at port (2)

$$= 20 \text{ mW} - 0.0454 \text{ mW} = \mathbf{19.9546 \text{ mW}}$$



Example 5.3 : A magic-T is terminated by a impedance of $Z_1 = 283.33 \, \Omega$ at port (1), by another impedance of $Z_2 = 116.67 \, \Omega$ at port (2) and some other impedance of $Z_4 = 150 \, \Omega$ at port (4), the E-arm. The H-arm (port-(3)) is connected to a $50 \, \Omega$ power source feeding 1 watt of power into it. Determine the power delivered to the 3 ports (1), (2) and (4) and the power reflected back by the junction which comes out of port (3).

Solution : Referring to fig. (5.15), an impedance of $Z_1 = 283.33 \, \Omega$ is connected to port (1).

$$\therefore \text{Reflection coefficient at port (1)} = K_1 = \frac{Z_1 - Z_0}{Z_1 + Z_0}$$

$$\therefore K_1 = \frac{283.33 - 50}{283.33 + 50} = 0.7$$

$K_1 = 0.7$

Similarly, an impedance of $Z_2 = 116.67 \, \Omega$ connected at port-(2) produces a reflection coefficient K_2 given by

$$\begin{aligned} K_2 &= \frac{Z_2 - Z_0}{Z_2 + Z_0} \\ &= \frac{116.67 - 50}{116.67 + 50} \end{aligned}$$

$$\therefore K_2 = 0.4$$

Impedance Z_4 connected at E-arm produces a reflection coefficient K_4 given by

$$\begin{aligned} K_4 &= \frac{Z_4 - Z_0}{Z_4 + Z_0} \\ &= \frac{150 - 50}{150 + 50} \end{aligned}$$

$$\therefore K_4 = 0.5$$

The $[S]$ matrix of magic-T is given by equation (5.85) as

$$[S]_{MT} = \begin{bmatrix} 0 & 0 & \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ 0 & 0 & \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} & 0 & 0 \\ \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} & 0 & 0 \end{bmatrix}$$

Let a_1 , a_2 , a_3 and a_4 be the normalized input voltages and b_1 , b_2 , b_3 and b_4 are the respective output voltages at ports 1, 2, 3 and 4.

Then

$$a_1 = K_1 b_1 = 0.7 b_1$$

$$a_2 = K_2 b_2 = 0.4 b_2$$

$$a_4 = K_4 b_4 = 0.5 b_4$$

Since 1 W of power is fed into H-arm (i.e., port-3), we have

$$P_{in} = P_3 = |a_3|^2 = 1 \text{ W}$$

$\therefore$

$$a_3 = 1 \text{ V}$$

We have

$$[b] = [S]_{MT} [a]$$

$$\begin{bmatrix} b_1 \\ b_2 \\ b_3 \\ b_4 \end{bmatrix} = \begin{bmatrix} 0 & 0 & \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ 0 & 0 & \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} & 0 & 0 \\ \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} & 0 & 0 \end{bmatrix} \begin{bmatrix} a_1 \\ a_2 \\ a_3 \\ a_4 \end{bmatrix}$$

$$\begin{bmatrix} b_1 \\ b_2 \\ b_3 \\ b_4 \end{bmatrix} = \begin{bmatrix} 0 & 0 & \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ 0 & 0 & \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} & 0 & 0 \\ \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} & 0 & 0 \end{bmatrix} \begin{bmatrix} 0.7 b_1 \\ 0.4 b_2 \\ 1 \\ 0.5 b_4 \end{bmatrix}$$

$$\therefore b_1 = \frac{1}{\sqrt{2}} [1 + 0.5 b_4] \quad \dots (5.121)$$

$$b_2 = \frac{1}{\sqrt{2}} [1 - 0.5 b_4] \quad \dots (5.122)$$

$$b_3 = \frac{1}{\sqrt{2}} [0.7 b_1 + 0.4 b_2] \quad \dots (5.123)$$

$$b_4 = \frac{1}{\sqrt{2}} [0.7 b_1 - 0.4 b_2] \quad \dots (5.124)$$

Substituting for b_1 and b_2 from equations (5.121) and (5.122) respectively into equation (5.124), we get

$$\begin{aligned} b_4 &= \left(\frac{1}{\sqrt{2}} \right) (0.7) \left(\frac{1}{\sqrt{2}} \right) (1 + 0.5 b_4) - \left(\frac{1}{\sqrt{2}} \right) (0.4) \left(\frac{1}{\sqrt{2}} \right) (1 - 0.5 b_4) \\ &= 0.35 (1 + 0.5 b_4) - 0.2 (1 - 0.5 b_4) \\ &= 0.35 + 0.175 b_4 - 0.2 + 0.1 b_4 \end{aligned}$$

$$\therefore b_4 = \frac{0.15}{1 - 0.175 - 0.1} = 0.2068$$

From equation (5.121),

$$b_1 = \frac{1}{\sqrt{2}} [1 + 0.5 b_4]$$
$$= \frac{1}{\sqrt{2}} [1 + 0.5 \times 0.2068]$$
$$\therefore b_1 = 0.78$$

From equation (5.122),

$$b_2 = \frac{1}{\sqrt{2}} [1 - 0.5 b_4]$$
$$= \frac{1}{\sqrt{2}} [1 - 0.5 \times 0.2068]$$
$$\therefore b_2 = 0.634$$

Substituting for b_1 and b_2 in equation (5.123), we get

$$b_3 = \frac{1}{\sqrt{2}} [0.7 \times 0.78 + 0.4 \times 0.634]$$
$$\therefore b_3 = 0.5654$$

$$\therefore \text{Power output at port (1)} = P_1 = |b_1|^2 = (0.78)^2 = 0.6084 \text{ W}$$

$$\text{Power output at port (2)} = P_2 = |b_2|^2 = (0.634)^2 = 0.4019 \text{ W}$$

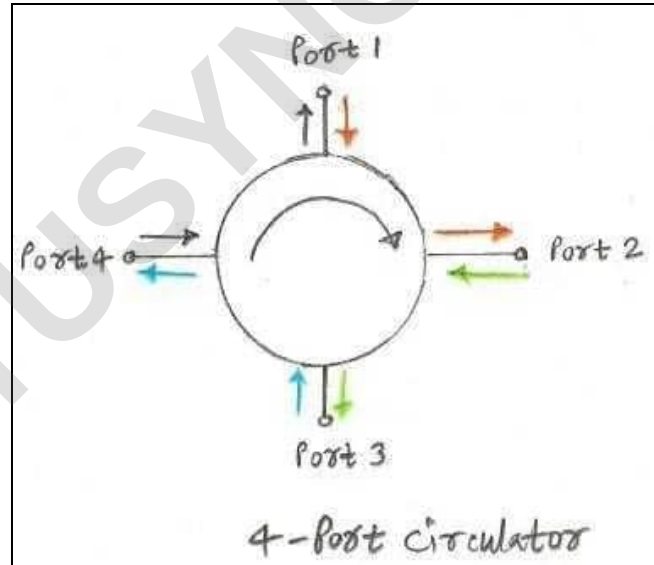
$$\text{Power output at port (4)} = P_4 = |b_4|^2 = (0.2068)^2 = 0.0427 \text{ W}$$

$$\text{Power reflected from junction coming to port (3)} = P_3 = |b_3|^2 = (0.5654)^2 = 0.3196 \text{ W}$$

Circulators

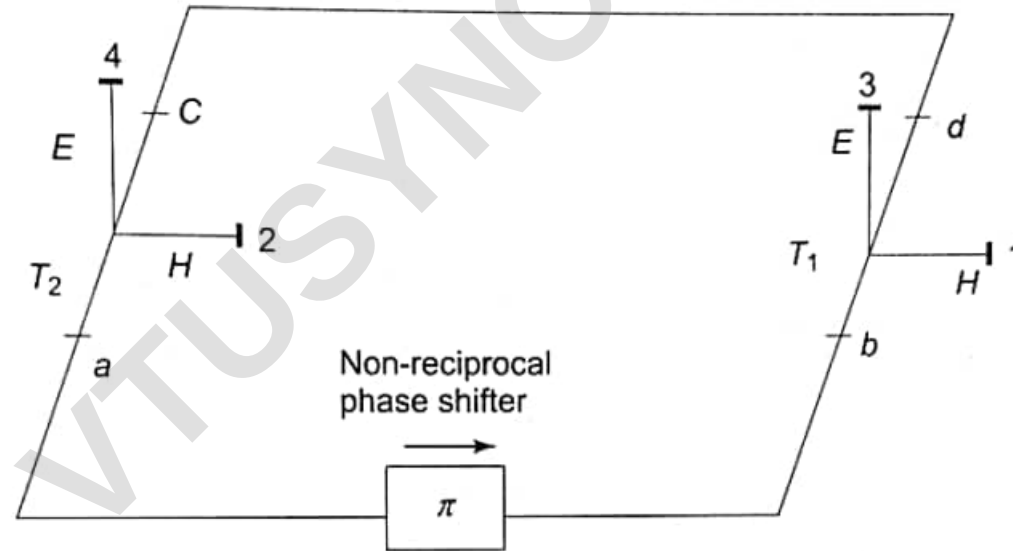
A circulator is a multiport junction in which the wave can travel from one port to the next port in one direction only.

Three-port & Four-port circulators are commonly used.



Four- port Circulator

A four-port circulator can be constructed from two magic Tees and a non-reciprocal 180 degree phase shifter.



Operation:

- **Input at port 1:** An input at port 1 is split into two in-phase and equal amplitude waves in the collinear arms 'b' and 'd' of the magic-tee T1 and added up to emerge from Port 2 in the magic-tee T2.
- **Input at port 2:** A signal at port 2 will split into two equal amplitude and equiphase waves in the collinear arms of the magic-tee T2 and appears at point 'b' and 'd' out of phase due to presence of non-reciprocal 180 degree phase shifter. These out of phase waves add up and appear from port 3 in the magic tee T1.

Similarly,

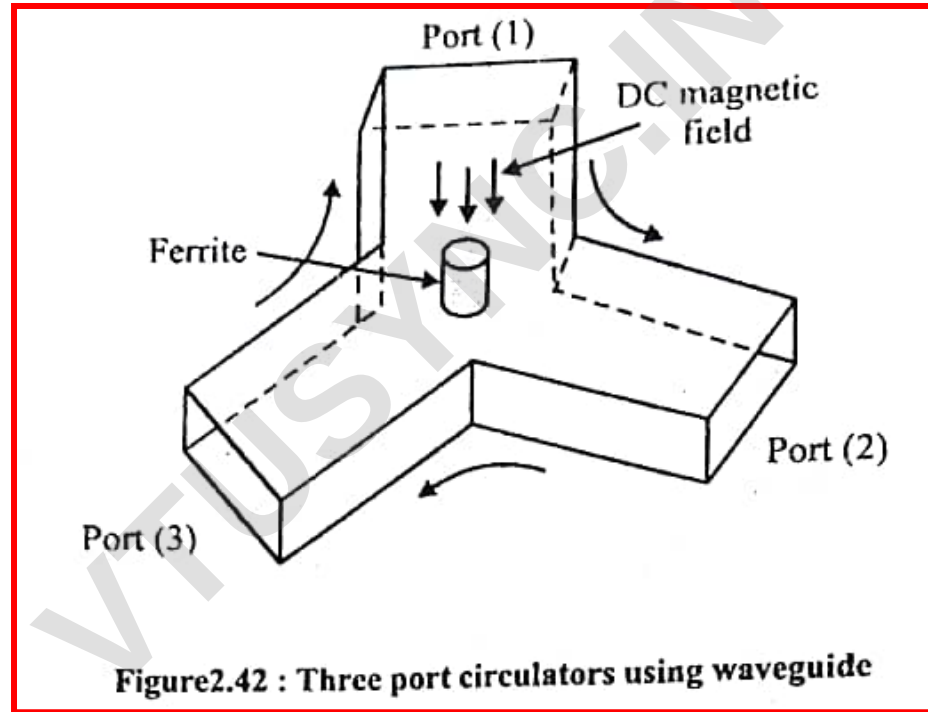
- An input signal at port 3 will emerge from port 4, and
- An input at port 4 will appear at Port 1.

Thus the circulator property is exhibited.

A perfectly matched, lossless, and non-reciprocal four-port circulator has S-matrix:

$$[S] = \begin{bmatrix} 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix}$$

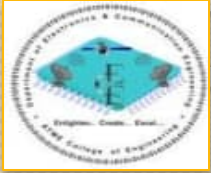
Three- port Circulator



(ii) **3-port Circulator:** 3-port circulator can also be constructed as shown in figure 2.42, by placing the ferrite exactly at the centre. The diameter and height of the ferrite and the strength of dc magnetic field are so designed to provide 120° Faraday rotation. Due to this the power fed at port (1) comes out of port (2) only, power into port (2) comes out of port (3) only and finally power into port (3) comes out of port (1) only.

3-port circulator is also sometimes called Y-circulator or Wye-circulator. As port (1) is connected to port (2), port (2) to port (3) and port (3) to port (1), we have the scattering coefficients $S_{12} = S_{23} = S_{31} = 1$. Thus for a perfectly matched, lossless, non-reciprocal Y-circulator, the scattering matrix is given by

$$S = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{bmatrix}$$

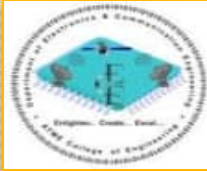


Isolators

Isolators

An "**isolator**" is a two-port non-reciprocal device which provides minimum attenuation to wave propagation in one direction and very high attenuation in the opposite direction. When such an isolator is inserted between the source and the load, almost the entire power is coupled to the load and the reflections from the load will be completely absorbed. Since no reflected power can enter into the source from the output side, the source remains stable with respect to its amplitude and frequency.

Faraday Rotation Isolators



(b) Faraday Rotation Isolator: Figure 2.44 shows a Faraday Rotation Isolator consisting of a piece of rectangular waveguide joined to a 45° twisted rectangular waveguide, then to a rectangular-to-circular transition, a circular waveguide, circular-to-rectangular transition and then to a rectangular waveguide. The rectangular waveguides at the two ends consists of ferrite cards whose plane is parallel to the broader dimension of the waveguide. A ferrite rod of suitable length tapered on both sides is placed co-axially with that of circular waveguide.

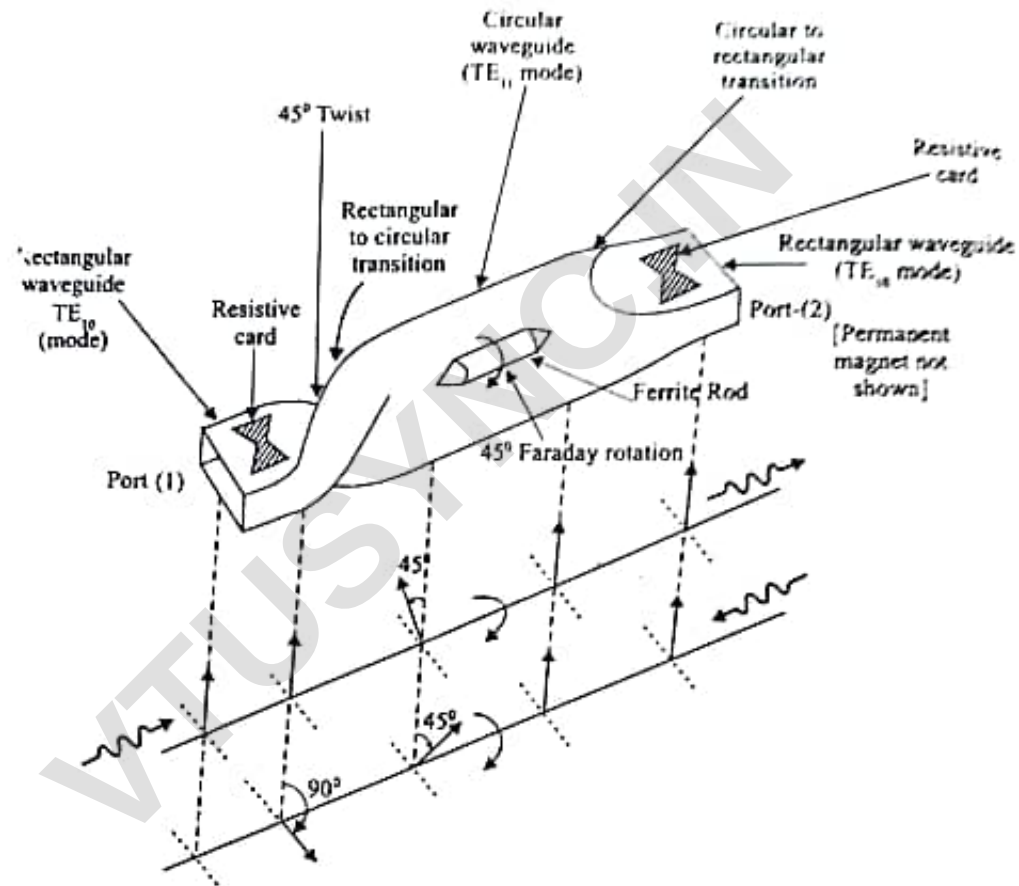
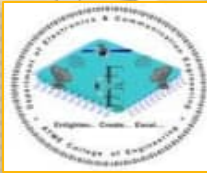


Figure 2.44 : Faraday Rotation Isolator

Operation of isolator : The resistive card which is placed along the broader dimension of the waveguide absorbs any wave whose plane of polarization is parallel to its own plane. The resistive card does not absorb any wave whose plane of polarization is perpendicular to its own plane.

A wave (TE_{10} mode) passing into port (1) gets shifted by 45° , because of twist, in anticlockwise direction. Because of ferrite, the plane of polarization gets rotated by 45° in clockwise direction and comes out of port (2) with minimum attenuation.

But a TE_{10} mode wave entering port (2) gets rotated by 45° due to Faraday rotation in the clock-wise direction (looking into port (1)) and rotates further by another 45° because of twisted waveguide. The plane of polarization is now in parallel with the resistive card, so that the wave is completely absorbed by the card. This power is dissipated in the resistive card as heat. In practice, 20 to 30 db isolation is obtained for transmission for port (2) to port (1) and insertion loss of about 1 db.



Thank You